

A unified single-cell atlas of human lung provides insights into pulmonary diseases



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Summary

Background The human lung is a highly complex organ characterised by extensive cellular heterogeneity, making it susceptible to a broad range of diseases. Single-cell transcriptomics has shed light on disease-specific cellular features, but previous studies have been fragmented, limiting a unified understanding of cellular mechanisms across various lung diseases. Furthermore, high-resolution reference atlases for lung cells are lacking, impeding the effective integration of spatial omics data and the exploration of shared pathogenic mechanisms.

Methods We constructed uniLUNG, the most comprehensive single-cell RNA sequencing atlas of the human lung, by integrating 62 published datasets comprising 9.2 million cells from 1807 donors across health and 17 disease conditions. We leveraged this high-resolution cell atlas to seek distinct cell types across diverse lung pathologies. By integrating spatial transcriptomics, we also identified transitional cell populations in lung cancer, and provided new insights into tumour evolution and the associated microenvironment.

Findings We present a comprehensive lung cell atlas, encompassing cellular data across major lung diseases and health states. Using this resource, we identified distinct cell populations, such as Lym-monocytes and T-like B cells, which are specifically enriched in certain lung diseases and linked to immune dysregulation. Furthermore, our spatially resolved multi-omics analysis revealed a transitional malignant subpopulation, NSCLC-like SCLC, which plays a key role in the transformation from non-small cell lung cancer (NSCLC) to small cell lung cancer (SCLC), driving tumour microenvironment remodelling.

Interpretation We offered a high-resolution, cross-disease lung cell reference that uncovers distinct cell types and cellular transitions critical to disease progression and therapeutic resistance. This resource provides essential insights into lung disease mechanisms and has important implications for the development of targeted therapeutic strategies, particularly in the context of lung cancer.

Funding This work was supported by the grants of National Key R&D Program of China (2021YFF1200900 and 2021YFF1200903), National Natural Science Foundation of China (92474107), Guangdong Basic and Applied Basic Research Foundation of China (2022B1515120077), Major Project of Guangzhou National Laboratory of China (GZNL2024A01003), and Support Scheme of Guangzhou for Leading Talents in Innovation and Entrepreneurship of China (2020007).

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Keywords: Single-cell transcriptome; Spatial transcriptome; Lung cell atlas; Multiple lung diseases; Lung cancers

Introduction

The lung, with its complex cellular ecosystem, poses significant challenges for precision medicine due to its pathological heterogeneity. Single-cell transcriptomics

has revealed cellular diversity in health and disease, but independent studies remain constrained by sample biases, technical variability, and disease-specific focuses, resulting in datasets with limited diversity.^{1–5}

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Research in Context

Evidence before this study

Lung diseases, including lung cancer, idiopathic pulmonary fibrosis, and chronic obstructive pulmonary disease, are marked by complex cellular heterogeneity. While single-cell RNA sequencing has provided valuable insights into the cellular diversity of the lung, studies have often been limited to disease-specific datasets and small sample sizes. There has been a lack of a comprehensive lung cell reference, which has hindered the interpretation of spatial omics data and limited the understanding of shared mechanisms across diverse lung conditions.

Added value of this study

This study integrates 62 published single-cell RNA sequencing datasets, totalling 9.2 million cells from 1807 donors, to construct uniLUNG, the most comprehensive lung single-cell atlas to date. The atlas includes a novel four-level cell annotation framework, which identifies 122 distinct lung cell types and provides a detailed cellular landscape of the

lung across different health and disease state as well as clues to distinct cell subtypes. By combining this resource with spatial transcriptomics, we identified key transitional cell populations, such as NSCLC-like SCLC, that drive histological transformation in lung cancer and significantly reshape the tumour microenvironment.

Implications of all the available evidence

The atlas provides a comprehensive, high-resolution lung cell reference that offers new insights into the cellular and molecular mechanisms underlying a range of lung diseases. By uncovering distinct cell types and transitional states involved in disease progression, this resource paves the way for future studies that will deepen our understanding of lung disease biology. Furthermore, it holds significant potential for advancing precision medicine, offering new avenues for targeted therapeutic strategies, particularly in lung cancer treatment and immune modulation.

Integrating multiple datasets into a comprehensive lung cell landscape can help overcome these limitations and provide deeper insights into lung biology. Meanwhile, spatial omics has enabled the reconstruction of molecular niches, tumour microenvironments, and novel molecular subtypes in lung diseases.^{6–8} However, its interpretation critically depends on high-resolution cell type information. By integrating single-cell and spatial data, researchers can better resolve spatial cell distributions, cell–cell interactions, and disease-associated subpopulations.^{9–11} Thus, a holistic lung cell landscape is essential for advancing spatial omics research in lung biology.

Recent studies have integrated diverse datasets to construct comprehensive cell atlases, revealing rare cell types and characterising disease-associated cellular heterogeneity.^{5,12–14} However, limitations in donor, dataset, sample, and cell type diversity, as well as metadata completeness,^{12,13,15,16} prevent these atlases from fully capturing lung cellular diversity. An ideal cell atlas should encompass exhaustive cellular, phenotypic, physiological, and multi-omics data across all cell types and disease states. The Human Lung Cell Atlas (HLCA),¹⁷ integrating data from healthy and diseased lungs, represents a milestone in lung biology. Yet, there is potential to enhance its data inclusivity, disease coverage, and cell diversity. Maximising the use of available datasets across as many diseases as possible can improve the representativeness of an integrated cell atlas and strengthen its role as a foundational reference.

In this study, we present uniLUNG (<https://lung.unifiedcellatlas.org>), the most comprehensive cell atlas of lung health and disease to date. It integrates datasets from individuals of diverse ethnicities, ages, and sexes,

encompassing 9.2 million cells across 18 lung states. Utilising a four-tiered Universal Hierarchical Annotation Framework (uHAF), uniLUNG classifies 122 known lung cell types, providing a detailed representation of lung biology in both health and disease.

Cross-disease analysis using uniLUNG identified two distinct cell subpopulations: Lym-monocytes and T-like B cells, which preferentially accumulate in specific lung diseases, revealing potential commonalities and unique features among these conditions. T-like B cells, enriched in LUAD and COVID-19 patients, may correlate with T-cell exhaustion, suggesting similar risks in CF, COPD, and ILD patients. Furthermore, by integrating spatial transcriptomics, we uncovered a key transitional malignant cell subpopulation (NSCLC-like SCLC) during the transformation of non-small cell lung cancer (NSCLC) to small cell lung cancer (SCLC). This subpopulation exhibits unique spatial gene expression patterns and drives NSCLC tumour microenvironment remodelling. This may serve as a signal for SCLC transformation in NSCLC patients.

Methods

Data collection and processing

To build uniLUNG, we collected cells from various research studies and public databases, including GEO, HCL, HCA, HLCA, DISCO, TISCH, LuCA, HTAN, CancerSEA, CancerSCEA, LungMAP, HuBMAP, UCSC Cell Browser, and Single Cell Expression Atlas.^{3,15,17–28} The datasets cover a diverse range of populations, including healthy individual and multiple lung diseases patients (Asthma, CF, COPD, COVID-19, CWP, ILD,

IPAF, IPF, NSIP, PF, LUAD, LUSC, LCC, SCLC, NSCLC, pneumonia and sarcoidosis). For each dataset, we collected the expression matrices and accompanying metadata, which includes information about sequencing technologies, donor age and sex, ethnicity, organ type, sample tissues, sampling site, disease status, smoking, covid and tumour stage, and original annotations. Regarding the ethnic information, this category is entirely derived from the original research and is reported by combining race, regional distribution and ethnicity, including Asian, European, Caucasian, African, American, African American, South Asian, Latino, East Asian, Hispanic, and Pacific Islander. In uniLUNG, we refer to this information as metadata.

To eliminate the impact of different preprocessing methods on data integration, we chose to only use datasets that provide the raw matrices to construct uniLUNG. Additionally, we used a list of 43,878 genes ([Supplementary Table S11](#)) approved by the HUGO Gene Nomenclature Committee (HGNC) to convert gene names from datasets that were originally named with Ensemble ID to HGNC symbols. Genes that were not detected in the sequencing were filled with zeros. Subsequently,

Based on previous studies^{4,17,29,30} and our own empirical testing, we constructed a standardised quality control pipeline suitable for most datasets. For each dataset, we excluded cells with fewer than 200 detected genes or with a high proportion of mitochondrial gene transcripts (>15%) and retained barcodes with a total transcript count between 500 and 40,000. The mitochondrial gene threshold was determined through testing across multiple core atlas datasets ([Supplementary Figure S2a](#)). DecontX³¹ was applied for environmental RNA correction, with a maximum allowable environmental contamination level set to 0.2. Subsequently, we selected 3000 highly variable genes (HVGs) and computed PCA based on 30 latent dimensions. Finally, we utilised DoubletFinder³² to identify and remove doublets from the dataset. We also evaluated the retention rate of cells before and after preprocessing across multiple datasets ([Supplementary Figure S2b](#)).

Benchmarking of data integration

Benchmarking for data integration methods was performed in uniLUNG core. It consists of 14 public datasets covering 11 lung states, sourced from 12 different sequencing platforms and technologies. To mitigate batch effects arising from diverse donor samples and sequencing technologies, we opted to partition large datasets encompassing multiple sequencing platforms and experimental designs. Ultimately, the uniLUNG core was subdivided into 34 independent datasets, representing batches requiring correction. We mapped the original labels to the

uHAF for consistent cell naming and the newly assigned label was utilised for supervised data integration tasks.

We selected the latest and widely used methods for benchmarking. These included methods implemented in Python such as CellHint,³³ Combat,³⁴ Harmony,³⁵ Scanorama,³⁶ scVI,³⁷ and scANVI,³⁸ as well as methods executed in R like BBKNN,³⁹ LIGER,⁴⁰ SeuratV4 (RPCA),²⁹ SeuratV5 (CCA, RPCA, and scVI),⁴¹ and FastMNN.⁴² Additionally, we conducted control tasks in both Python and R using only PCA-based computations, labelled as PCA and SeuratV5_PCA, respectively, to assess performance without employing any specific integration methods. Some methods could be run with different parameters or versions, such as CellHint and CellHint_harmonize, SeuratV4_RPCA and SeuratV5_RPCA. Furthermore, certain methods were already integrated into SeuratV5, including SeuratV5_FastMNN and SeuratV5_Harmony. We treated these integrated methods as new tasks for individual testing purposes. Due to memory usage constraints, the CCA method based on SeuratV4 was excluded from the evaluation. For all tasks run in R, SeuratV4 was utilised by default for methods not explicitly specified. The conversion of dataset formats between R and Python was facilitated by SeuratDisk.

All integration tools used in this work were conducted on 3000 highly variable genes (HVGs) with default parameters. These HVGs were selected to ensure consistent processing across methodologies. Among all strategies, scANVI and scVI utilise the original counts of HVGs as input, while the other methods use scaled matrices normalised with default parameters. Both scANVI and CellHint_harmonize methods permit guidance of data integration through known cell type labels. We mapped the original labels to the uHAF for consistent cell naming. Following this mapping, each cell's original label was assigned a corresponding cell type in the uHAF, aiming for higher resolution whenever possible. This newly assigned label was employed for supervised data integration tasks.

In the benchmarking, apart from cell-type LISI (cLISI) and integration LISI (iLISI), other metrics (NMI, ARI, Cell type ASW, Graph connectivity, PCR comparison, and Batch ASW) were computed using the default parameters of the scIB⁴³ framework. cLISI and iLISI were calculated using the R package LISI³⁵ within the Python environment. Specifically, cLISI was used to assess clustering accuracy, while iLISI evaluated integration performance. The LISI scores range from 1 to N (the number of batches). For iLISI, higher values indicate better mixing (1 = complete separation), whereas for cLISI, lower values reflect better biological variance preservation (1 = ideal re-clustering). To ensure interpretability, all LISI scores were normalised to a 0–1

scale, where higher values denote better performance. The calculation formula is as follows:

$$iLISI_{scale} = \frac{Median(cLISI) - Min(cLISI)}{Max(cLISI) - Min(cLISI)}$$

$$cLISI_{scale} = \frac{Min(cLISI) - Median(cLISI)}{Max(cLISI) - Min(cLISI)}$$

We evaluated the quality of each integration using eight metrics, with four assessing batch correction quality and four quantifying the preservation of biological variance post-integration. Batch correction and biological conservation scores were calculated as the respective averages of the corresponding metrics. The overall score was then determined by a weighted average of 0.4:0.6 for batch correction and biological conservation scores, respectively.

Integration and manual cell type re-annotation of uniLUNG core

Based on benchmark results, we opted for scANVI with the highest overall score for integration of the uniLUNG core. It was executed using the unified original cell labels named through uHAF and default parameters. Following data integration, we computed the nearest neighbour graph based on 30 latent dimensions with neighbours set to $k = 30$. Subsequently, the graph underwent the first round of Leiden clustering at a resolution of 0.1. Leveraging uHAF, we assigned primary level labels to all cells, encompassing endothelial cells, epithelial cells, fibroblasts, lymphocytes, myeloid cells, and cancer cells, with cancer cells retaining their original annotations.

Subsequently, except for cancer cells, each cell subtype underwent computation of a new nearest neighbour graph using scANVI's 30 latent dimensions, with k set to 30. The new graph underwent a second round of Leiden clustering at a resolution of 0.2. Based on the clustering outcomes, each cell subtype was further delineated, completing the second-level annotation, and the same strategy was employed for the third and fourth rounds of Leiden clustering. Specifically, during each iteration of clustering annotation, we conducted manual quality checks on each cluster to exclude potential doublet clusters or dead/dying cell clusters.

Cell type label transfer from the uniLUNG core to extension

After completing integration and annotation of the uniLUNG core, we used CellTypist⁴⁴ to transfer its labels to the uniLUNG extension, following best practices for large-scale label transfer (<https://celldtypist.readthedocs.io>). We first sampled up to 20,000 cells per cell type (mode = 'each', n_cells = 20,000, balance_cell_type = True) and performed a rapid model pretraining (n_jobs = 10, max_iter = 10, use_SGD = True). From these initial

models, we extracted the top 300 marker genes per cell type based on absolute regression coefficients and combined them across types for final model training. For downstream training, we used logistic regression (max_iter = 100) to obtain unbiased probability estimates.

Two models were trained from the uniLUNG core: one including cancer cells (used for lung cancer datasets), and one excluding them (used for all other datasets). Label transfer to the uniLUNG extension was performed using default settings with majority_voting = True enabled. The resulting cell annotations in the extension provide a robust reference for user-specific cell type identification and supervised data integration based on uniLUNG. Both CellTypist models are available on our atlas website (model_all.pkl and model_noca.pkl in <https://lung.unifiedcellatlas.org/download/>).

It is important to note that the labels in the extended atlas are exclusively derived through CellTypist-based label transfer without subsequent manual verification. Therefore, these automated annotations should not be directly used for downstream analytical applications without further validation. When utilising the extended atlas data, researchers should implement appropriate data integration strategies aligned with specific research objectives, and perform rigorous manual cell type annotation through established biological validation methods.

Data standardisation and database architecture

During preprocessing, we standardised gene expression matrices for each dataset to ensure consistency across genes. Existing batch effect correction methods are typically tailored for specific downstream analysis tasks, highlighting the importance of selecting appropriate correction methods based on research questions and analytical objectives. Therefore, we normalised gene expression values across all datasets solely based on library size to address differences caused by sequencing depth, ensuring comparability of gene expression values across batches. Other sources of variance were preserved along with the expression matrix. Results obtained from the online platform and downloaded data undergo only normalisation, with users required to select suitable methods for data integration to mitigate batch effect influences in further analyses. Additionally, we provide pre-integrated atlases for direct download and downstream analysis, including the uniLUNG core and five sub-atlases.

All expression data and metadata are stored in a unified format using the uGT, a column-based NoSQL database deployed in cloud storage. Unlike traditional SQL systems, uGT supports high-dimensional data retrieval, enabling efficient storage of single-cell profiles with 40,000–50,000 features. Each cell is stored as a row: columns 1–43,878 contain gene expression values, while metadata occupy subsequent columns.

Web portal construction and data availability in uniLUNG

We developed a graphical user interface portal (<https://lung.unifiedcellatlas.org>) to facilitate user access to uniLUNG. The portal is built using the JavaScript-based Vue framework, with statistical data visualised via the ECharts plugin. To support diverse user needs and promote uniLUNG as a reference for future single-cell lung studies, we provide multiple data access options through a dedicated download interface. Users can obtain pre-integrated resources such as the core atlas and sub-atlases for direct downstream use, or filter cells based on custom conditions to retrieve corresponding expression data. Notably, filtered data are normalised for sequencing depth but not batch-corrected, requiring users to perform their own integration as needed. Comprehensive data, including all expression matrices and detailed uHAF information, are also available in the download section (<https://lung.unifiedcellatlas.org/#/uHAFItems>).

Atlas analysis for multiple lung diseases

Data integration and main cell type annotation

We used Harmony³⁵ for data integration while removing batch effects between datasets. Following integration, we computed the nearest neighbour graph using 3000 highly variable genes based on 30 principal components, with neighbours set to $k = 30$. After an initial round of Leiden clustering at a resolution of 0.1, all cells were classified into the following categories based on uHAF: endothelial cells, epithelial cell, fibroblasts, lymphocytes, and myeloid cell. Subsequently, we performed a second round of clustering on lymphocytes and myeloid cells at a resolution of 0.2, leading to the identification of macrophage, mast cell, monocyte, B cell, NK cell, and T cell.

Cell abundance calculation

In the analysis of lung health and multiple diseases, significant disparities exist among datasets representing different lung statuses, primarily stemming from imbalances in donor numbers, sample sizes, and cell quantities. These discrepancies prevent us from directly comparing changes in cell proportions across different lung states to obtain accurate results. Therefore, we opted to employ scCODA⁴⁵ to compute and compare changes in cell abundance. Cell abundance reflects the extent of change for each cell type relative to a reference cell, which remains relatively stable across different lung statuses. In scCODA, all samples were stratified according to sample type and sampling site, with sequencing platform and individual donors as covariates. The abundance differences of cell types were calculated separately within each sample layer. We used the automatically calculated results from scCODA as the reference cell type (reference_cell_type = "automatic"), but the sample layers with insufficient sample numbers and

the inability to automatically calculate baseline cell types were not compared. Additionally, we set the false discovery rate (FDR) to 0.05.

Data sampling of monocytes and B cells

We extracted monocyte and B cell subsets from the raw data, excluding samples with low cell counts, resulting in 108,813 monocytes and 244,857 B cells. Batch effects were corrected using Harmony, followed by unsupervised clustering at a resolution of 0.01. To accommodate downstream computational constraints while preserving the original embedding structure, we applied Seurat's SketchData function with the LeverageScore method for uniform subsampling. This yielded ~11,000 monocytes (~10%) and ~12,500 B cells (~5%) for subsequent analysis.

NMF analysis of monocytes and B cells

We performed non-negative matrix factorisation (NMF) on the sampled monocyte and B cell populations using the R package NMF,⁴⁶ with log-transformed expression matrices as input. The analysis used the standard "brunet" method over 50 iterations, testing rank values from 2 to 8. The optimal rank was determined by the largest change in cophenetic correlation across ranks, yielding a rank of 3 for monocytes and 4 for B cells.

Cell type annotation and AUCell score calculation of monocyte and B cell subpopulations

Based on the Leiden clustering and NMF results, monocytes were categorised into CD16 monocyte, CD14 monocytes, and lymphocyte-like monocyte. B cells were divided into memory B cell, activated B cells, and T cell-like B cells. We then combined lymphocyte characteristic gene sets (CD79B, CD19, PTPRC, GZMA, CD8A, CD3D, IL7R, CXCR4, NKG7, CD8B) and T cell characteristic gene sets (CD3E, CD3G, CD8A, CXCL13, GATA3, CD3D, IL7R, IL2RB, CD8B, CD4) to score gene modules for both monocyte and B cell subpopulations.

Odds ratios (OR) calculation

According to the method of Zheng et al.,⁴⁷ the distribution preference of monocytes and B cell subsets in different pulmonary states was calculated.

Differential expression analysis

We first applied the R package Libra⁴⁸ to aggregate single-cell counts into pseudo-bulk profiles for each sample. Differential expression analysis was performed using edgeR⁴⁹ with a generalised linear model and likelihood ratio test, incorporating donor ID as a covariate to account for batch effects (model.matrix (~donor_ID + group), where group denotes cell subtypes). In parallel, we used Seurat's FindMarkers function to identify differentially expressed genes for Lym-monocytes and T-like B cells. The final gene list

consisted of the intersection of results from both edgeR and Seurat.

Enrichment analysis

We conducted GO enrichment analysis using clusterProfiler.⁵⁰ The parameters of pvalueCutoff and qvalueCutoff were both set to 0.01.

Correlation analysis of signatures

The signatures of T-like B cell (top 30) and Exhausted T cell (HAVCR2, TIGIT, LAG3, PDCD1, CXCL13, TIM3, PD1, LAYN)^{51,52} were selected as gene set, and LUAD tumour samples and normal tissue samples in TCGA were selected for correlation analysis based on GEPIA2.⁵²

Transcription factor analysis by SCENIC

We applied SCENIC⁵³ to analyse active transcription factors in monocytes and B cells. The co-expression networks were constructed, and then the potential target genes of each transcription factor were identified by GRNBoost based on co-expression. DNA-motif analysis was conducted using RcisTarget to detect potential transcription factor binding motifs. And the regulatory activity of each transcription factor in each cell was scored by AUCell.

Reference query mapping for Lym-monocytes and T-like B cells

We used scArches⁵⁴ to map Lym-monocytes and T-like B cells to reference datasets. HLCA and LuCA were selected as references for Lym-monocytes, while pan-cancer B and T cell atlases^{47,55} were additionally included for T-like B cells. To address disparities in cell numbers between the query and reference datasets, all reference datasets were uniformly down-sampled to 200,000 cells, excluding any with fewer cells. These references were used to train an scVI model with 3000 highly variable genes over 500 iterations. The same gene set was extracted from the Lym-monocyte and T-like B cell subsets for query model training, also with 500 iterations. Finally, the reference and query datasets were aligned in a shared embedding space using the trained model.

Atlas analysis for lung cancers

CNV score estimation

We utilised the R package inferCNV⁵⁶ to compute copy number variation (CNV) events across all epithelial cells, employing a cutoff of 0.1. Epithelial cells from healthy lung tissues within the uniLUNG core were chosen as the reference. To quantify the occurrence of CNV events across all regions within cells, we normalised the gene expression levels of each cell to a range of -1 to 1 as described by Peng et al.⁵⁷ Then, we computed the squared sum of the normalised values as the CNV score, indicating the extent of CNV events within each cell across all regions.

Evaluation of cell differentiation potential

For each cell type or cell subpopulation, CytoTRACE⁵⁸ was used to evaluate its differentiation potential, generating a UMAP visualisation coloured by age group, as well as a boxplot representing the overall differentiation potential across different cell groups.

Trajectory construction

Pseudotime analysis was conducted using Monocle3⁵⁹ to investigate the differentiation relationship among all malignant cells from LUAD, LUSC and SCLC patient. Gene exhibiting significant changes over pseudotime were identified through the graph_test function, with q value < 1e-4. We used the plot_genes_branched_pseudotime function to plot differential gene expressions.

Survival analysis

LUAD samples in TCGA were selected for survival analysis based on GEPIA2.⁵²

Preprocessing and integration of spatial transcriptomics data

We performed basic filtering and normalisation of spatial transcriptomics data from tumour samples of LUAD and LUSC patients using Scanpy.⁶⁰ The filtering criteria included a maximum count of 50,000, less than 20% mitochondrial counts, and a minimum of 50 cells per gene. Based on 3000 highly variable genes, we applied the SEDR⁶¹ algorithm specifically developed for spatial transcriptomics to compute the embedding layer, which was then utilised for data integration via Harmony.

Identification of spatial regions

We utilised the mclust_R function in SEDR to cluster the LUAD and LUSC datasets, while employing the Cancer-Finder⁶² algorithm to identify malignant spots. Based on clustering results and the identification of malignant cells, we categorised all spots into normal, pre-malignant, and malignant regions, with consideration of EPCAM expression levels.

Analysis of spatial gene expression patterns and ligand-receptor co-localisation

To assess the widespread impact of the NSCLC-like SCLC subset on LUAD and LUSC tumours, we employed SpaGene⁶³ to identify gene expression patterns in samples exhibiting high NSCLC-like SCLC characteristics, setting nPattern to 10. We then used the SpaGene_LR function to evaluate ligand-receptor co-localisation in each sample, filtering for shared ligand-receptor pairs between LUAD and LUSC samples.

Identification of spatial communication signal flow

We analysed communication signal flow in samples with and without high NSCLC-like SCLC characteristics in LUAD and LUSC using COMMOT.⁶⁴ All procedures were conducted following the official guidelines.

Spatial mapping of cell types using cell2location

We selected LUAD and LUSC samples from LuCA as reference datasets. After removing all mitochondrial genes, we applied the `filter_genes` function for further filtering (`cell_count_cutoff` = 5, `cell_percentage_cutoff2` = 0.03, `nonz_mean_cutoff` = 1.12) to prepare for spatial mapping and cell type annotation. The filtered reference dataset was then used to train a regression model, with ‘cell_type_major’ and ‘dataset’ as batch and label variables, respectively, and ‘platform’ as a covariate. The regression model was trained on all cells in the dataset with default parameters, achieving convergence within 250 epochs.

For spatial mapping, we set the expected number of cells per spot to 30 and RNA sensitivity regularisation parameter to 20 (default). Training on the full LUAD and LUSC datasets converged after 40,000 iterations. We extracted spatial cell abundances using the 5th percentile of the posterior distribution and computed cell type-specific gene expression at each spatial location to support NCEM-based cell–cell communication analysis.

Non-negative matrix factorisation based on spatial mapping

Building on the cell2location mapping results, we employed non-negative matrix factorisation (NMF) to further identify spatial co-localisation of cell types. After training within a factor range of 3–10, we selected 6 factors for cell compartment identification and visualisation in the LUAD and LUSC datasets.

Analysis of intercellular dependency as a spot composition function

We used the NCEM method⁶⁵ to assess the influence of spot/ecological composition on all genes. To focus on biologically relevant genes, we selected gene sets described in WikiPathways from the molecular signature database (MSigDB), yielding 6716 genes. All 24 cell types in LuCA’s ‘cell_type_major’ variable were included in the NCEM analysis. Following type coupling analysis on the filtered dataset, we further dissected these couplings based on specific gene effects, including the impact of all senders on a single receiver (receiver effect analysis) and the influence of a single sender on all receivers (sender effect analysis). We retained dependencies involving at least 500 differentially expressed genes, with a *q*-value < 0.05 and absolute log fold change >1 (1.5 for LUSC). Finally, sender similarity analysis was performed to characterise the sender profiles of CD4 T cells in LUAD patients and cDC2 cells in LUSC patients under different conditions.

Statistics

The statistical tests used in this study include the *t*-test (implemented in Scanpy) for identifying marker genes in each cell cluster, the Wilcoxon rank-sum test (the default method in Seurat) and the likelihood ratio test

(implemented in edgeR) for differential expression analysis, as well as a Bayesian model (implemented in scCODA) for compositional analysis. This study is entirely based on publicly available datasets, and the sample sizes and data filtering criteria are specified in the manuscript. No blinding was applied, as all analyses were performed on de-identified datasets and the investigators had no influence on group assignments or outcome assessments.

Ethics

This work is part of a joint project led by Tsinghua University with collaborators such as Sun Yat-sen University. The ethical approval has been granted by the Ethics Committee of Zhongshan School of Medicine of Sun Yat-sen University (approval reference number: 2024-061). This work incorporated publicly accessible and anonymised data; thus, written informed consent was deemed not necessary.

Role of funders

The funders did not play any role in the study design, data collection, management, analysis, interpretation, writing, review, approval of the manuscript, or the decision to submit the manuscript for publication.

Results

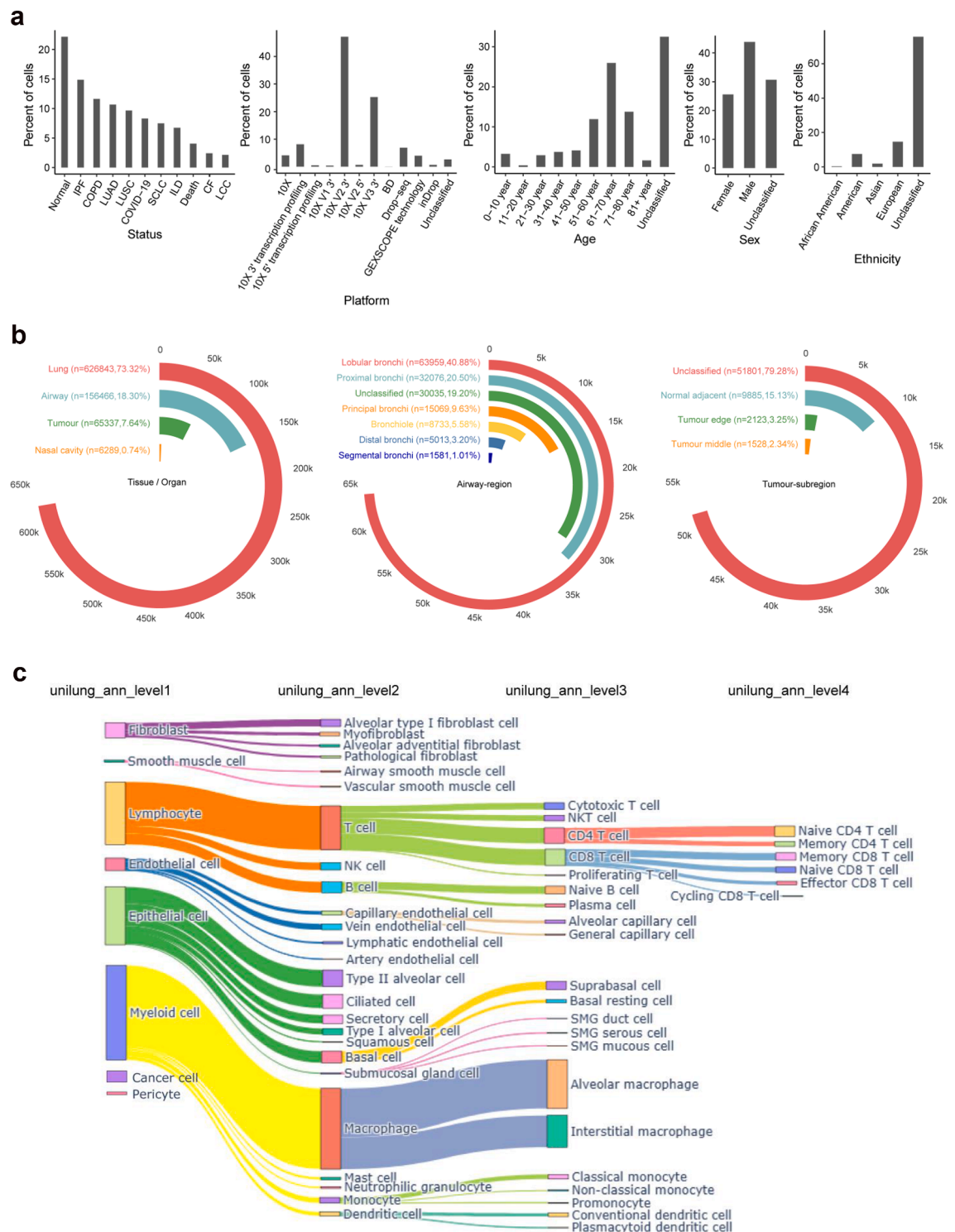
The construction the of uniLUNG through supervised integration

The representativeness of uniLUNG core datasets

We first constructed the core atlas and expanded it into a complete atlas by transforming labels to address computational challenges associated with processing millions of cells. The uniLUNG core was developed from 14 public datasets, encompassing 311 donors across 11 lung conditions, including healthy, cystic fibrosis (CF), chronic obstructive pulmonary disease (COPD), COVID-19, interstitial lung disease (ILD), idiopathic pulmonary fibrosis (IPF), large cell carcinoma (LCC), LUAD, LUSC, and SCLC (Supplementary Table S1). To ensure representativeness, we balanced cell counts across conditions and prioritised datasets with complete original cell annotations and comprehensive metadata, covering donor demographics (age, sex, ethnicity), disease status, smoking history, and sample-specific details such as tissue type, sampling site, experimental protocol, and sequencing platform (Fig. 1a and b). Key disease-related information, including COVID-19 severity and tumour staging, was also retained.

A four-level universal hierarchical annotation framework (uHAF) in uniLUNG

Current single-cell lung studies employ diverse cell classification standards, complicating cross-study comparisons. A large-scale integrative atlas offers a valuable



framework for standardised cell type annotation across datasets. Based on extensive existing research and consensus such as HLCA,^{17,66,67} particularly the insights from LungMAP/CellRef,³ we manually developed a Universal Hierarchical Annotation Framework (uHAF) for human lung cell-level annotation (Supplementary Figure S3), encompassing 122 known lung cell types (Supplementary Table S2). At the same time, we strive to unify the anatomical framework in the original research into a coherent framework and map them onto the ontology (Supplementary Figure S4 and Supplementary Table S3).

Data integration benchmarking optimises embedding

After uniformly preprocessing all datasets, we obtained 854,935 cells to establish the uniLUNG core. To effectively mitigate batch effects while preserving biological differences, we evaluated 18 data integration strategies, encompassing 11 major and latest methods of data integration^{29,33–42} (see Methods). For supervised integration methods requiring known cell type labels, we created uniform identity labels by mapping original cell annotations to the uHAF (Supplementary Table S4). We benchmarked these strategies using scIB⁴³ with eight metrics assessing batch correction and biological conservation (Supplementary Figure S5a). Based on these evaluations, we selected the top-performing method scANVI to construct the uniLUNG core.

Manual cell type re-annotation achieves maximum cell diversity with high accuracy

Following data integration, we systematically re-annotated all cells in the uniLUNG core based on the uHAF. The core was initially classified into five major cell types along with malignant cells. Through an iterative annotation process, we ultimately identified 60 distinct cell types within the core atlas (Figs. 1c and 2a, and Supplementary Figure S5b). For each cell type, we carefully examined the expression of canonical marker genes during the iterative refinement (Fig. 2b and Supplementary Figure S6). In addition, we compared the re-annotated results with the original labels at both the first and second hierarchical levels to assess consistency (Fig. 2c).

The creation of complete uniLUNG via label transfer from the core atlas

Based on the uniLUNG core, we utilised CellTypist⁴⁴ to expand the consensus annotation of cells to other datasets through label transfer (see Methods), resulting in a comprehensive transcriptomic atlas of human lung cells spanning healthy and 17 disease states (uniLUNG: <https://lung.unifiedcellatlas.org>, see Methods and Supplementary Table S5). This atlas integrates 62 published single-cell datasets, encompassing 9,201,161 cells from 58 cell types obtained from 1807 donors, along with biological and demographic metadata (age, sex, ethnic,

disease status, and sampling region). uniLUNG's construction relies on a unified framework including high-quality preprocessing, uHAF- and CellTypist-based annotations, assembly into the unified giant table (uGT),⁶⁸ and a user-friendly data portal. Standardised gene expression and metadata integration in uGT enables cell-centric analyses and cross-condition gene expression comparisons.

uniLUNG reveals rare cell types across health and multiple lung diseases

Identification of two distinct cell subpopulations across multiple diseases via uniLUNG

The human lung is vulnerable to diverse diseases, yet our understanding of its cellular diversity across health and disease remains limited. Most existing single-cell studies focus on specific conditions due to constraints in sample type and availability, lacking a comprehensive, cross-disease perspective. Leveraging uniLUNG, we analysed 1,643,385 cells from tissue samples spanning 18 lung conditions (Supplementary Table S6), providing an unprecedented resource for single-cell research in pulmonary disease. After batch correcting with Harmony,³⁵ cells were categorised into five major cell types (Fig. 3a and b). We confirmed that cell-type labels assigned in the core atlas were largely retained after re-integration into the extended atlas, supporting the robustness of our annotation across embeddings (Supplementary Figure S7a). Given the heterogeneity in sample types, donor numbers, and cell counts across different disease datasets, we stratified samples by tissue type and anatomical location and applied scCODA⁴⁵ within each stratum to assess differences in cell-type composition (excluding malignant cells). This revealed significant shifts in epithelial, lymphoid, and myeloid populations across disease states within multiple sample strata (Fig. 3c, Supplementary Figure S8). To further reduce confounding from epithelial malignancy, we subdivided immune populations into T cells, B cells, NK cells, macrophages, mast cells, and monocytes and repeated the analysis (Supplementary Figure S9a). This revealed disease-associated changes in the abundance of macrophages, monocytes, B cells, and T cells (Supplementary Figures S9b and S10).

As monocytes and B cells remain underexplored in current lung disease research, we applied NMF and Louvain clustering to these two lineages and identified three distinct subpopulations within each (Fig. 3d, Supplementary Figure S11 and S12a). Validation of these subpopulations was performed through the expression analysis of canonical markers for myeloid, monocyte, lymphocyte, and B cell lineages (Supplementary Figure S12b). Strikingly, we identified two distinct subsets: a monocyte subpopulation exhibiting elevated expression of lymphocyte markers (termed Lym-monocytes) and a B cell subpopulation enriched for T cell markers (termed T-like B cells). Both subpopulations

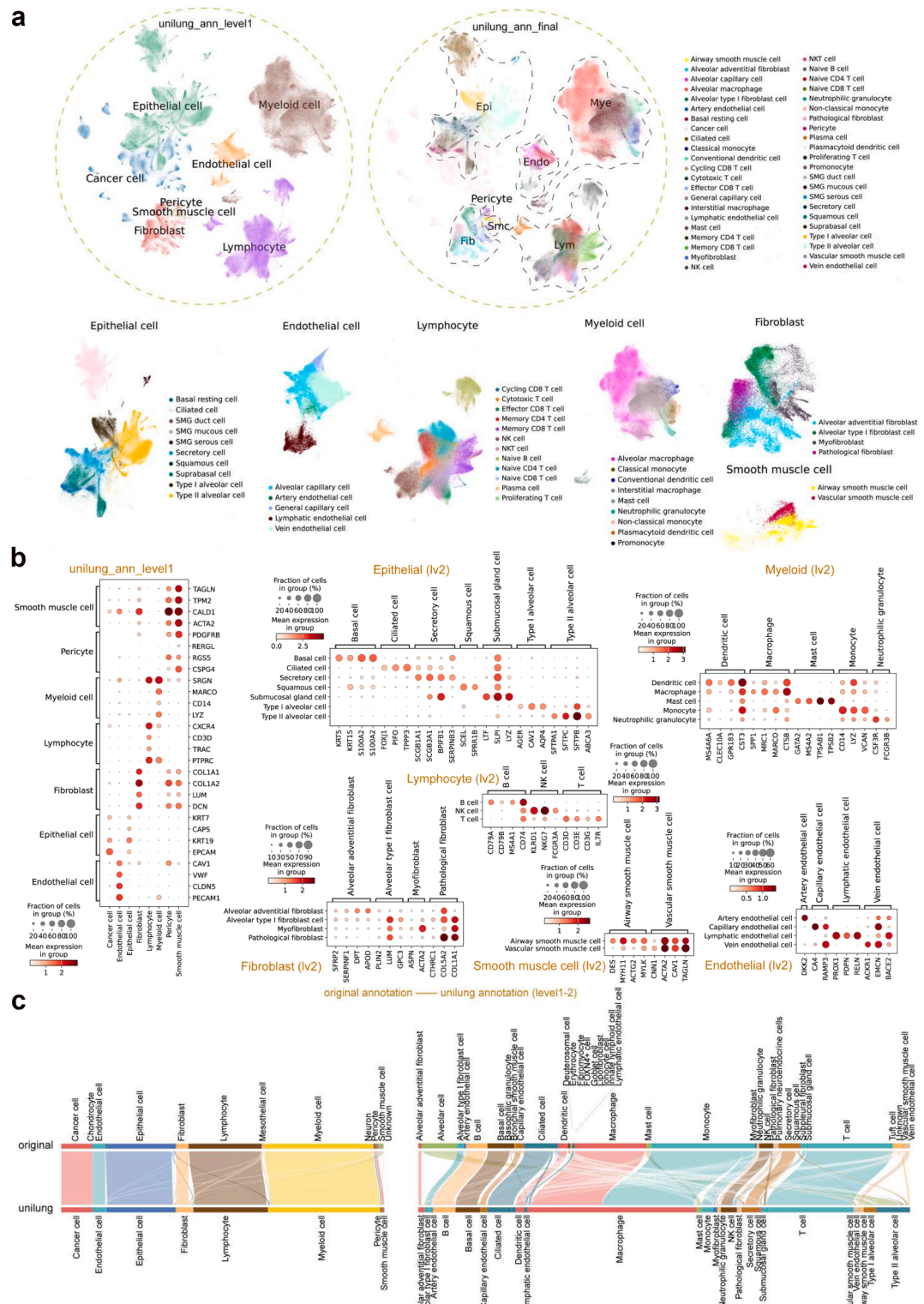


Fig. 2: unilUNG core achieved a reliable re-annotation based on uHAF. (a) UMAP of the unilUNG core coloured by the re-annotation results at the first and final level. **(b)** Expression of marker genes for cell types at the first and second levels in the core atlas. **(c)** Comparison between the re-annotation and the original labels of unilUNG core reveals good consistency at the first and second level of annotation.

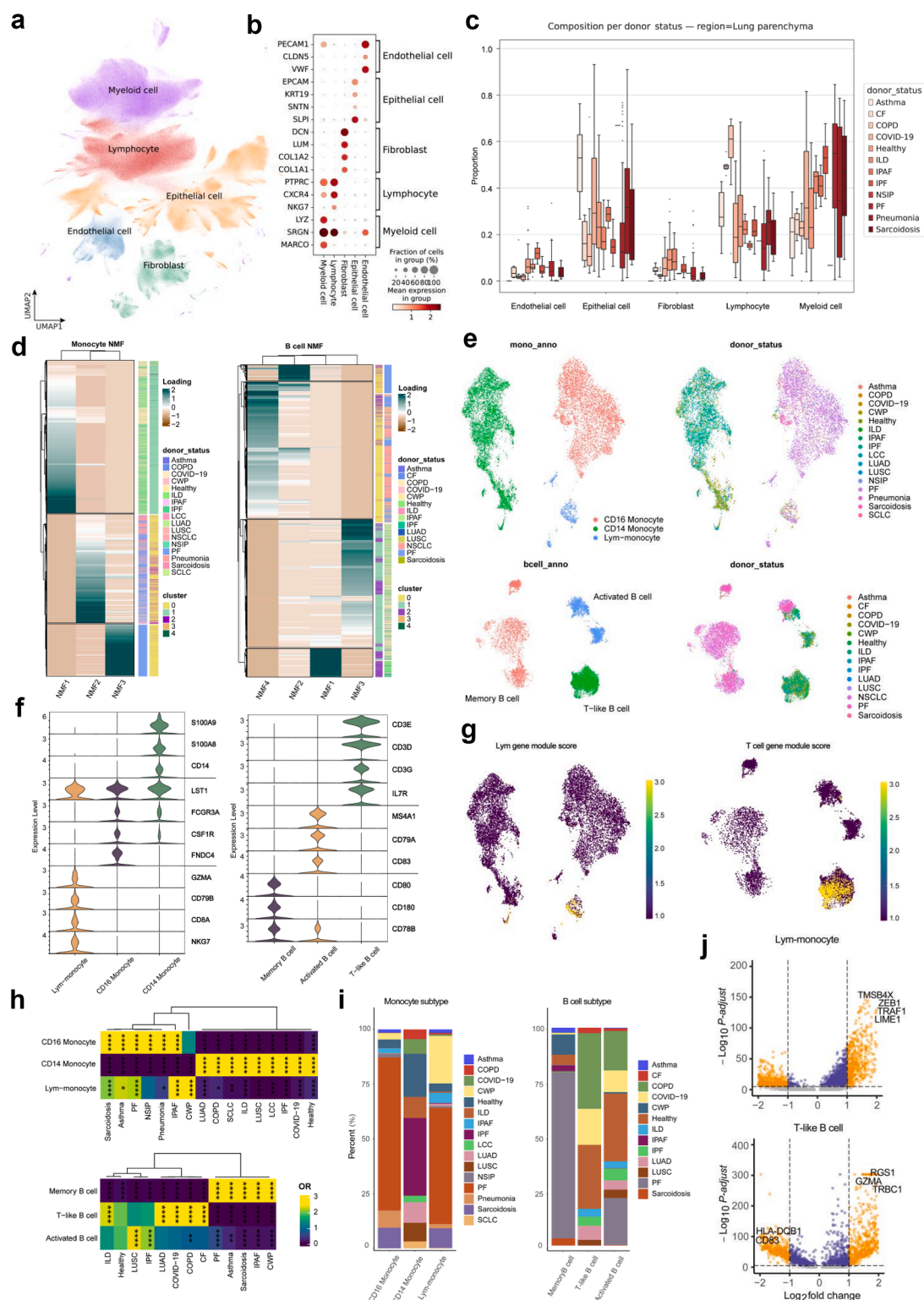


Fig. 3: Analysis of lung health and multiple diseases. (a–b) Identification of major cell types. (c) Data represent the abundance of main cell type in lung parenchyma samples across multiple lung disease conditions. The central line indicates the median, boxes denote the inter-quartile range (IQR), and whiskers extend to $1.5 \times$ IQR. (d) NMF analysis divided B cells and monocytes into 3 and 4 subpopulations,

were reproducibly detected across diverse lung disease cohorts (Fig. 3e and f). Importantly, we rigorously excluded potential confounding factors arising from doublets during the uniLUNG preprocessing. In addition, we performed cross-validation using multiple doublet detection algorithms under varying assumed doublet rates across all monocyte and B cell clusters (Supplementary Figures S13 and S14), further ruling out the possibility that Lym-monocytes and T-like B cells represent artefactual doublets. Gene set enrichment analysis revealed that lymphocyte signature genes and T cell signature genes were highly enriched in Lym-monocytes and T-like B cells, respectively (Fig. 3g).

Heterogeneity and characterisation of the two distinct cell subpopulations

To further dissect monocyte and B cell heterogeneity across lung diseases, we calculated odds ratios (OR)⁴⁷ and cell proportions, highlighting disease-specific enrichment of subtypes such as Lym-monocytes and T-like B cells (Fig. 3h and i). CD14 monocytes were broadly and consistently distributed, while CD16 monocytes showed more restricted patterns, largely excluding CD14 cells but overlapping with Lym-monocytes in PF, IPAF, and CWP. Similarly, memory B cells were enriched in PF, IPAF, and CWP, whereas T-like B cells were more widely distributed across COPD, COVID-19, LUAD, ILD, and other conditions.

Differential expression analysis revealed key marker distinctions: Lym-monocytes and T-like upregulated lymphocyte markers (e.g., TMSB4X, ZEB1, TRAF1, LIME1), while T-like B cells expressed T cell markers (e.g., RGS1, GZMA, TRBC1) and downregulation of some B cell markers (e.g., CD83, HLA-DQB1) (Fig. 3j). GO enrichment further confirmed that Lym-monocytes were enriched for lymphocyte- and T cell-related processes, while T-like B cells were associated with mitochondrial respiration and energy metabolism pathways (Fig. 4a).

T-like B cells were significantly more abundant in severe/critical COVID-19 patients compared to mild/moderate cases (Fig. 4b). Given that T cell exhaustion in severe COVID-19⁶⁹ is linked to impaired mitochondrial function and oxidative stress,⁵¹ the rise of T-like B cells may represent a compensatory mechanism. Additionally, the T-like B cell signature in LUAD tumour was highly correlated with an exhausted T cells gene program (Fig. 4c), suggesting potential remodelling of the tumour microenvironment (TME) in response to T cell exhaustion in tumour,⁷⁰ and leading to a similar functional compensation mechanism.

We further investigated regulatory programs using SCENIC. In monocytes, FOS was broadly active, including in Lym-monocytes, where ETS showed specific upregulation (Fig. 4d and Supplementary Figure S15a). Their targets included genes such as FCN1, PTPRC, MKI67, and SERPINA1 (Supplementary Figure S15b), with ETS-linked targets enriched for lymphocyte-associated markers (Fig. 4e). Moreover, Transcriptional networks also differed markedly between CD14, CD16, and Lym-monocytes (Supplementary Figure S15). In B cells, each subpopulation exhibited distinct transcription factor activity. Notably, TFDP1 and KLF6 were highly active in T-like B cells, regulating T cell-associated markers such as IL7R and CTSA (Supplementary Figure S16).

Validation of rare cell subpopulations with external scRNA and CITE-seq datasets

To confirm the presence of Lym-monocytes and T-like B cells, we used scArches⁵⁴ to map them onto other existing lung atlases. In the HLCA,¹⁷ Lym-monocytes mapped close to non-classical monocytes, reinforcing their monocyte-like characteristics, while T-like B cells clustered with CD4 and CD8 T cells, reflecting their similarity to T cells (Fig. 4f and g). Given their existence in lung cancer samples, we further mapped them using the LuCA.²³ Lym-monocytes aligned with classical monocytes and macrophages, while T-like B cells localised near CD4, CD8, and regulatory T cells (Supplementary Figure S17a and b). Projecting T-like B cells onto a published pan-cancer B⁵⁵ and T cell atlas⁴⁷ further confirmed their presence in lung cancer datasets, where they overlapped partially with B cells but did not correspond to any known cancer-associated B cell subtype, and were recognised as a distinct subpopulation despite their similarity to CD4/CD8 T cells (Fig. 4h and i). To further validate T-like B cells, we applied a reference-mapping approach to a published CITE-seq dataset (<https://data-browser.lungmap.net/projects/834e0d16-71b6-4425-a8ab-022b5000961c>) of COVID-19 lung. Consistent with our previous finding of their enrichment in COVID-19 lungs, T-like B cells were also identified in this dataset (Supplementary Figure S17c). Moreover, the surface protein data confirmed that T-like B cells exhibited a mixed profile of T- and B-cell markers, yet remained distinct from conventional B- and T-cell subsets (Supplementary Figure S18). To further assist interpretation, we additionally compared the expression of key T- and B-cell marker genes across three B cells subsets and T cells (Supplementary

respectively. (e) UMAP visualisation of B cell and monocyte subpopulations. (f) Characteristic gene expression in B cell and monocyte subpopulations. (g) Lymphocyte feature gene set score in monocytes and T-cell feature gene set score in B cells showed lymphocyte feature of Lym-monocyte and T cell feature of T-like B cell, respectively. (h) Distribution bias of B cells and monocyte subpopulations in different lung states. (i) The percentage of B cells and monocyte subpopulations derived from different lung states. (j) Gene differential analysis of T-like B cells and Lym-monocytes.

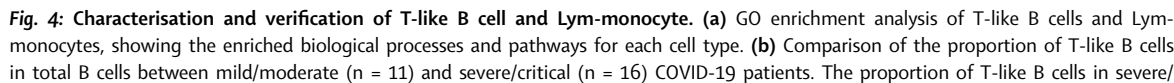


Figure S19). This provides independent evidence supporting their identity as a unique population. Together, these results provide robust transcriptomic and proteomic evidence that T-like B cells and Lym-monocytes represent distinct and reproducible subpopulations not prominently captured in previous lung atlases, underscoring the unique power of uniLUNG to uncover the diversity of lung cell types.

In summary, uniLUNG unveils the dynamic cellular landscape of the human lung across health and disease, particularly highlighting the heterogeneity of B cells and monocytes. The identification of two distinct cell types (Lym-monocytes and T-like B cells) reveals their distinct characteristics and their preference for specific lung conditions, showcasing the unparalleled scope of uniLUNG in enhancing our understanding of lung cell diversity and cross-disease insights.

Spatially resolved multi-omics deciphers NSCLC-to-SCLC transition

Lung cancer remains the leading cause of cancer-related deaths worldwide and is broadly classified into small cell lung cancer (SCLC) and non-small cell lung cancer (NSCLC), the latter including lung adenocarcinoma (LUAD), lung squamous cell carcinoma (LUSC), and large cell carcinoma (LCC). LUAD, LUSC, and SCLC represent the most common types, accounting for 80–85% (LUAD and LUSC) and 10–15% (SCLC) of all lung cancer cases (<https://www.cancer.org/cancer/types/lung-cancer>). Typically, lung cancer subtypes originate from distinct cell types and exhibit similar clinical features. However, emerging evidence suggests that histological transformations can occur between subtypes, such as NSCLC-to-SCLC^{71–73} and LUAD-to-LUSC.⁷⁴ These transformations alter tumour characteristics and promote drug resistance and poor prognosis,^{74–76} posing challenges for treatment development. Although prior studies have identified key drivers and microenvironmental changes associated with these transformations, the spatial dynamics remains poorly understood, despite the critical role of spatial architecture in tumour biology.^{77,78} To address this gap, we integrated uniLUNG with spatial transcriptomics data to investigate the remodelling of the tumour microenvironment during histological transformation, aiming to identify key factors and their roles in spatial microenvironmental reorganisation.

Cross-diseases atlas reveals transitional NSCLC-SCLC state of malignant cells

We analysed 290, 89, and 21 LUAD/LUSC/SCLC patients from the lung cancer sub-atlas of uniLUNG, respectively, part of whom had tumour staging information (Supplementary Table S7). Initially, all 1,683,056 cells were categorised into five main cell types in uHAF, with 310,555 epithelial cells further identified as basal cells, ciliated cells, secretory cells, AT1 cells, AT2 cells, and malignant cells (Fig. 5a–c). Consistent with our first case study, we observed that cell-type labels from the core atlas were largely preserved after re-integration into this lung cancer atlas, supporting the robustness of the annotation (Supplementary Figure S7b). Malignant cell identification was based on comprehensive marker gene expression, CNV scores, and cell differentiation potential (Fig. 5c, Supplementary Figure S20).

Clustering of all malignant cells revealed similarities between cancer cells from LUSC and LUAD, the two main NSCLC subtypes, while SCLC malignant cells formed a distinct cluster (Fig. 5d), highlighting significant biological differences between SCLC and NSCLC. Previous studies suggest that histological transitions between lung cancer subtypes may contribute to resistance to targeted therapies.^{71–75} To investigate potential lineage transitions among lung cancer subtypes, we performed trajectory analysis on malignant cells, which were grouped into four distinct branches (Fig. 5e, top right). The largest branch (Partition 1) encompassed malignant cells from LUAD, LUSC, and SCLC, forming a dominant trajectory (Fig. 5e, top left). By integrating tumour staging information for cells along this trajectory and for the separate SCLC cluster, we inferred a directionality of progression (Fig. 5e, bottom), which revealed a potential evolutionary path from NSCLC to SCLC. This suggests that a subset of malignant cells in NSCLC patients may evolve into SCLC within the tumour microenvironment (Fig. 5e, bottom).

We further investigated the expression dynamics of known NSCLC-SCLC transition drivers along the trajectory (Fig. 5f and Supplementary Figure S21a). Key markers showed consistent patterns: RB1, typically inactivated in SCLC,⁷³ and the EGFR family (EGFR, ERBB2, ERBB3), which are highly expressed in well-differentiated alveolar type II cells but downregulated in transformed SCLC cells,⁷⁶ decreased towards the

critical patients was significantly higher than that in mild/moderate patients (p-value = 0.0299). (c) Correlation analysis between exhausted T cells and T-like B cell gene signatures, performed separately in normal and tumour samples of LUAD. The results showed that the T-like B cell signatures were significantly correlated with the T cell exhaustion signatures in LUAD tumour samples ($R = 0.9$). (d) Differential activation of transcription factors across different monocyte subsets. (e) Transcription factor regulatory network centred on ETS1, which specifically activated in Lym-monocyte, highlighting its lymphoid characteristics. (f) UMAP visualisation of Lym-monocytes mapped to the monocyte region within the HLCA. (g) UMAP visualisation of T-like B cells mapped to the HLCA, showing the mapping to the CD4 T and CD8 T cell regions, with neighbouring B cells highlighted. (h) UMAP visualisation of T-like B cells mapped to the pan-cancer B cell atlas, revealing their proximity to, but distinct from existing B cell subpopulations. (i) UMAP visualisation of T-like B cells mapped to the pan-cancer T cell atlas. Unsupervised clustering shows T-like B cells primarily cluster in a distinct group, with surrounding cell types highlighted.

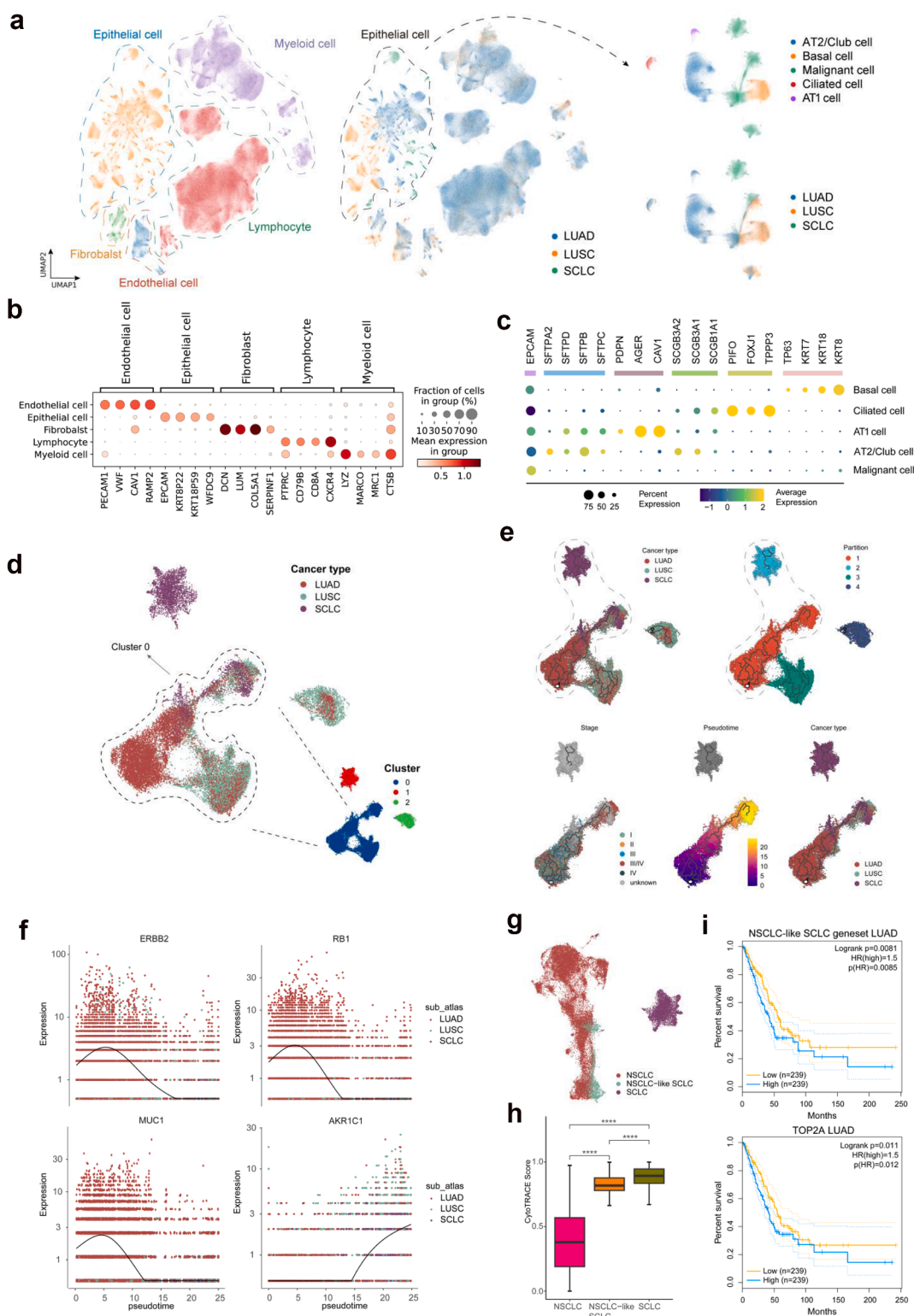


Fig. 5: Single cell transcriptome analysis of lung cancer subtype transition. (a–b) All cells were initially classified into five major cell types, with epithelial cells further subdivided into five distinct subpopulations. (c) Marker gene expression analysis of epithelial cell subpopulations, showing that the malignant cell subset expresses few markers of normal epithelial subpopulations, except for EPCAM. (d) UMAP visualisation

trajectory end. MUC19, overexpressed in LUAD and associated with tumour progression and metastasis,⁷⁹ has been shown at the single-cell level to inversely correlate with SCLC transformation and act as a key regulatory factor in LUAD-to-SCLC transformation,⁸⁰ which also showed progressive downregulation. In contrast, AKR1C1 is associated with acquired EGFR-TKI resistance in LUAD patients,⁸¹ and a similar resistance mechanism may be gained during the NSCLC-to-SCLC transformation.⁷² Additionally, genes related to proliferation, such as MKI67, TOP2A, and TPX2, also showed increasing expression along the inferred temporal progression^{82–84} (Supplementary Figure S21b).

We then separated all cells along the trajectory (Partition 1) and the independent SCLC subpopulation (Partition 2), and rebuilt their UMAP (Fig. 5g). A distinct subset of SCLC malignant cells present on the trajectory was designated as NSCLC-like SCLC to distinguish it from the independent SCLC population (Fig. 5g). Notably, this subpopulation exhibited a differentiation potential intermediate between NSCLC and SCLC cells, suggesting that it represents a transitional state during NSCLC to SCLC progression (Fig. 5h). Survival analysis based on the NSCLC-like SCLC gene signatures and TOP2A (a trajectory-associated gene also enriched in this signature) in TCGA LUAD samples revealed a significant association with poor prognosis in TCGA LUAD patients (Fig. 5i).

Spatial characterisation of NSCLC-like SCLC cells in NSCLC patients

We integrated spatial transcriptomic data from 12 tumour samples of 5 LUAD patients and 8 tumour samples of 3 LUSC patients.⁸⁵ Tumour, tumour-like, and normal regions were delineated based on unsupervised clustering, CancerFinder⁶² malignancy inference, and EPCAM expression (Fig. 6a and b). In each sample, EPCAM expression was spatially concentrated in regions identified as tumour and tumour-like cells (Fig. 6c, Supplementary Figure S22). Scoring each

sample with the NSCLC-like SCLC signature gene set revealed tumour cells with similar characteristics in some LUAD and LUSC tumour regions, confirming the presence of regions (highlighted in yellow) that express the NSCLC-like SCLC characteristics in NSCLC (Fig. 6d, Supplementary Figure S23).

In samples exhibiting high NSCLC-like SCLC scores, we identified several gene expression patterns (Supplementary Figure S24a). Most notably, we detected consistent enrichment of cancer-associated fibroblast (CAF) markers (Pattern 6) across LUAD and LUSC samples (Supplementary Figure S24a), suggesting widespread CAF infiltration. Basal cell markers (Pattern 4) and type II alveolar/club cell markers (Pattern 9) were enriched in LUSC and LUAD (Supplementary Figure S24b), respectively, reflecting their distinct malignant origins.^{86,87} Spatial colocalisation analysis revealed that ECM ligands (COL1A1, COL1A2, FN1, LUM)⁸⁸ co-localise with ITGB1, a receptor previously shown to be upregulated in SCLC⁸⁹ and implicated in EMT activation,⁹⁰ in tumour-adjacent normal regions (Supplementary Figure S25 and Supplementary Table S8). These results indicate that CAF-derived ECM remodelling and ECM-ITGB1 signalling may form a key component of the transitional microenvironment. To further dissect signalling dynamics, we applied COMMOT⁶⁴ to analyse intercellular communication in tumour samples with high and low NSCLC-like SCLC scores across LUAD and LUSC (Supplementary Table S9 and S10). In LUAD samples with pronounced NSCLC-like SCLC features, we observed directional MIF signalling from malignant to adjacent normal regions (as indicated by the black arrows), consistent with its role in driving immune inflammation⁹¹ and tumour progression (Fig. 6e). In LUSC, the MK pathway exhibited similar directional flow, aligning with its known function in promoting epithelial-mesenchymal transition (EMT) and tumour proliferation⁹² (Fig. 6e). Together, these findings suggest that the spatially variable signalling environment differs by subtype, potentially guiding divergent mechanisms of malignant transition.

of all malignant cells, coloured by cancer type. Unsupervised clustering revealed a cluster containing malignant cells from LUAD, LUSC, and SCLC patients, suggesting shared characteristics between malignant cells from different subtypes of lung cancer. (e) Trajectory analysis of all malignant cells identified four independent trajectories, distributed across different partitions. The largest partition encompasses trajectories representing malignant cells from LUAD, LUSC, and SCLC patients, which were established as the main trajectory. Based on clinical information, the direction of this main trajectory was determined to follow NSCLC-SCLC. (f) Driver factors showing significant expression changes along the trajectory. (g) UMAP visualisation was reconstructed by extracting all cells from the main trajectory and an independent SCLC malignant cell subpopulation. A small subset of SCLC patient-derived malignant cells within the main trajectory displayed similarities to NSCLC malignant cells, with differences from the independent SCLC malignant cell subset, and were named the NSCLC-like SCLC subpopulation. (h) The differentiation potential of the NSCLC-like SCLC subpopulation lies between the NSCLC and SCLC malignant cell subpopulation (p-value < 0.0001). Boxplots show the median, interquartile range (IQR), and whiskers extending to 1.5 × IQR (not error bars). (i) Survival analysis was performed on LUAD samples from TCGA based on the NSCLC-like SCLC signature gene set (p-value = 0.0085) and TOP2A expression (p-value = 0.012). The results indicate that the NSCLC-like SCLC subpopulation is associated with poor prognosis in LUAD patients.

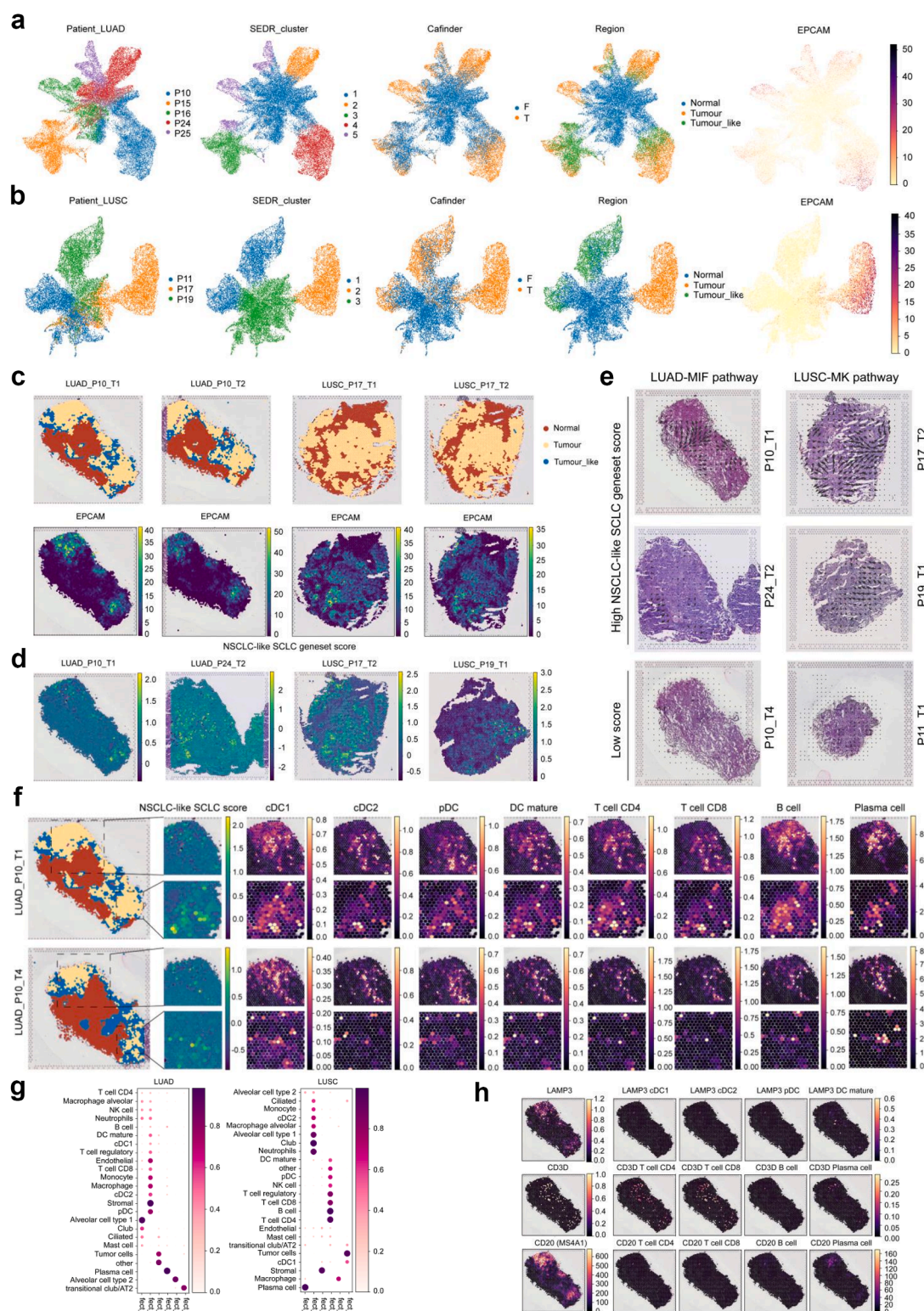


Fig. 6: Spatial characterisation of NSCLC-like SCLC subpopulation. (a–b) Spatial transcriptomic samples of LUAD (n = 12) and LUSC (n = 8) were integrated separately. Based on clustering results from SEDR and CancerFinder identification of malignant cells, all cells were classified into tumour cells, normal cells, and tumour-like cells, with EPCAM expression tested. (c) Spatial partitioning of regions containing tumour

NSCLC-like SCLC is associated with spatial immune microenvironment remodelling

Using the LuCA as a reference, we applied cell2location⁹³ to deconvolute cell types in LUAD and LUSC samples. The analysis showed increased abundance of cDC1, cDC2, and pDC in LUSC tumour regions, whereas LUAD tumours were enriched with cDC1, cDC2, pDC, DC mature, T cell CD4, T cell CD8, B cells and plasma cells (Supplementary Figure S26–S29). Notably, plasma cells were concentrated within the tumour in LUAD, while in LUSC, they predominantly resided in the surrounding normal tissue (Supplementary Figure S30a). Additionally, the presence of the NSCLC-like SCLC subpopulation significantly elevated immune cell abundance in both LUAD and LUSC (Fig. 6f, Supplementary Figure S30b). NMF analysis further confirmed distinct immune infiltration patterns between these tumour types (Fig. 6g). Importantly, lymphocytes enriched at the tumour core or margin in LUAD were identified as key components of tertiary lymphoid structures (TLS).^{94,95} Marker gene analysis confirmed the presence of TLS-associated cell types, such as DC-LAMP (LAMP3), CD3D-expressing T cells (CD3D), and B cells (CD20)^{96,97} (Fig. 6h). This suggests that the regions showing NSCLC-like SCLC characteristics may be associated with the formation of TLS in LUAD, potentially creating an immunologically dynamic niche.

By analysing the relationship between spatial spot composition and gene expression variability within cell types, we found that the NSCLC-like SCLC subpopulation significantly reshapes intercellular communication in LUAD and LUSC (Fig. 7a, Supplementary Figure S31). In LUAD samples with high NSCLC-like SCLC characteristics, CD4 T cells showed a loss of dependency on type II alveolar cells, tumour cells, and transitional club/AT2 cells, instead relying on cDC1 cells, indicating a shift in communication partners (Fig. 7a). Additionally, while mast cells and monocytes were critical communication hubs in both LUAD groups, their intercellular dependencies were significantly reduced in the low-scoring group (Fig. 7a). In LUSC, cDC2 cells in the high-scoring group exhibited strong dependencies on macrophages and stromal cells, whereas these dependencies were nearly absent in the low-scoring group (Supplementary Figure S31). We then performed receptor effect analysis to highlight

how sending cells influence gene expression in specific receptor cell types, providing context for differential gene expression in certain cellular couplings (Fig. 7b, Supplementary Figure S32). For CD4 T cells and cDC2 cells, which exhibited significant changes in intercellular dependency in LUAD and LUSC, we assessed the impact of the NSCLC-like SCLC subpopulation on these interactions through sender similarity analysis. We found that LUAD samples with lower NSCLC-like SCLC scores retained sender profiles of specific lineages more effectively (Fig. 7b), whereas the opposite was observed in LUSC (Supplementary Figure S32). Finally, given the distinct interactions between tumour cells and other cell types in the LUAD group (Fig. 7a), we further analysed how tumour cells, as single senders, affect gene expression in different NSCLC-like SCLC scoring contexts (Supplementary Figure S33).

In summary, by integrating the lung cancer sub-atlas of uniLUNG with spatial transcriptome, we reconstructed the malignant cell transition trajectories among LUAD, LUSC, and SCLC, identifying a distinct malignant cell subpopulation with NSCLC-like features—termed NSCLC-like SCLC. This subpopulation plays a critical role in reshaping the spatial tumour microenvironment in NSCLC, potentially representing a key transitional state during the progression from NSCLC to SCLC. It exhibits unique spatial gene expression patterns and intercellular signalling networks, while differentially influencing immune cell infiltration and cell type dependencies in LUAD and LUSC patients. These findings unveil the spatial landscape of histological transformation in lung cancer, providing new insights into the underlying mechanisms and potential risk factors for SCLC transformation in NSCLC patients. This case study underscores the broad potential of generating novel insights into lung diseases by integrating single-cell atlases with spatial omics data.

Discussion

From a methodological perspective, we established a nearly 10-million-cell cross-disease and health reference atlas for the human lung, complemented by a powerful data retrieval and comparison platform. Our meticulously designed cell type annotation framework, uHAF, along with a comprehensive benchmarking of 18 data integration strategies, establishes a new standard and

cells, normal cells, and tumour-like cells in LUAD and LUSC samples. (d) Scoring of NSCLC-like SCLC characteristic gene sets in LUAD and LUSC samples. Results revealed NSCLC-like SCLC characteristics (highlighted in yellow) in the tumour cell regions of some LUAD and LUSC samples, suggesting. (e) LUAD and LUSC samples with or without prominent NSCLC-like SCLC features were grouped into high and low scoring categories. Spatial communication analysis of signal flow in these groups indicated that in both LUAD and LUSC high-scoring samples, signals related to tumour progression were observed flowing from tumour cell regions to surrounding normal regions. (f) Two tissue sections from the same LUAD patient exhibited varying levels of NSCLC-like SCLC features, with the sample showing prominent NSCLC-like SCLC features displaying a more pronounced Immune cell infiltration in the tumour cell regions. (g) NMF analysis revealed the clustering of multiple immune cells at the same factor level in both LUAD and LUSC. (h) Marker expression of key cell types TLS in LUAD samples with prominent NSCLC-like SCLC features suggests the presence of TLS in these samples.

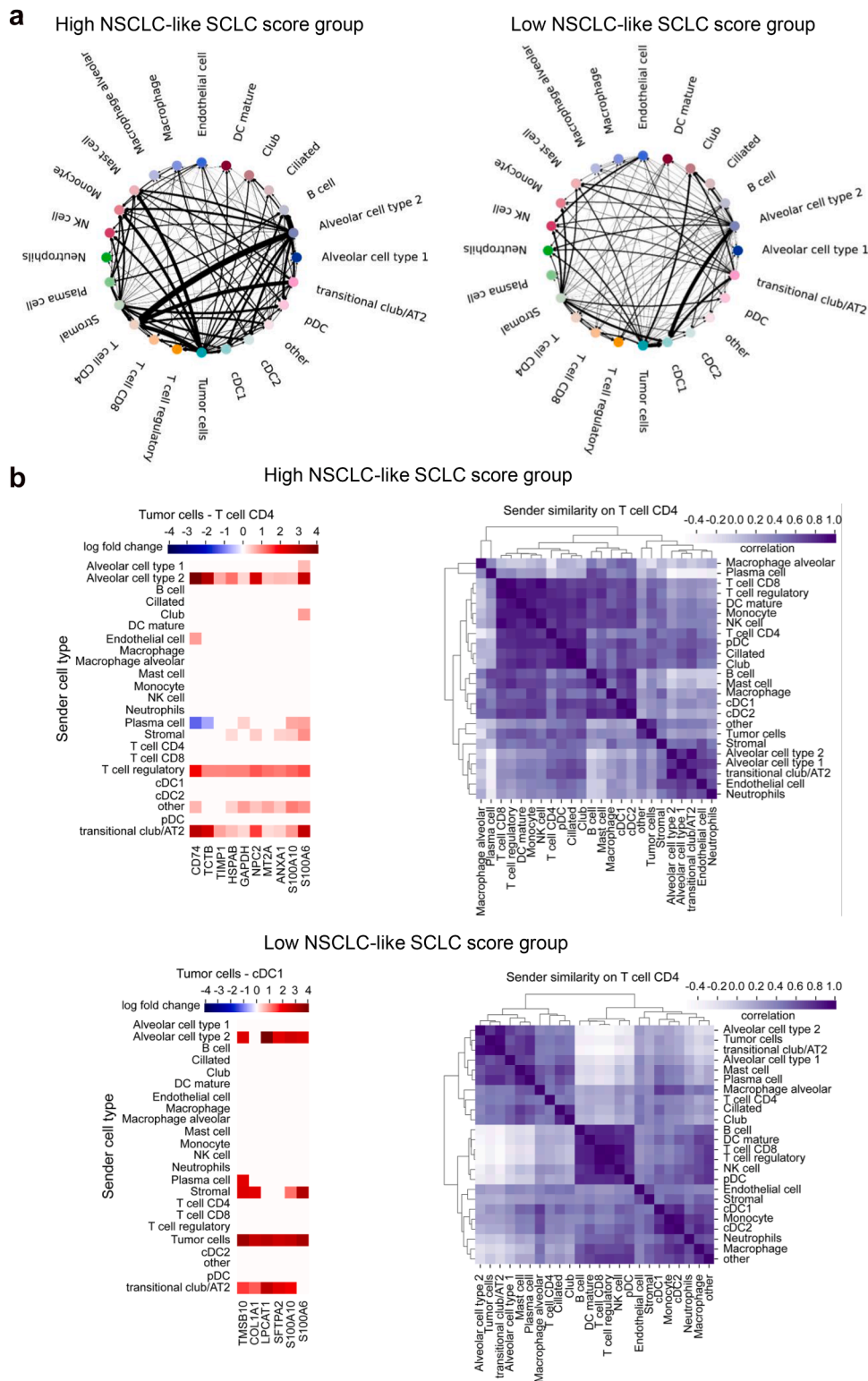


Fig. 7: Reshapes of intercellular dependence and sender effect in LUAD and LUSC tumour microenvironment by NSCLC-like SCLC cell. (a) Type coupling analysis with edges weighted by directional dependency strength, derived from fold changes of differentially expressed genes in sender-receiver cell type pairs across NSCLC-like SCLC score-stratified groups. The effect of the presence of NSCLC-like SCLC

guidance for large-scale cell atlas development. From a scientific and medical standpoint, we unveiled distinct cell types shared across lung diseases, uncovering potential common pathological alterations, such as T cell exhaustion, that may underlie diverse pulmonary conditions. Furthermore, by integrating spatial transcriptomics data, we identified a key transitional malignant cell subpopulation during lung cancer histological transformation and elucidated its role in reshaping the tumour spatial microenvironment. These findings vividly illustrate the immense value of a single-cell atlas that fully captures lung cellular diversity, not only in advancing single-cell transcriptomics research but also in driving the emerging field of spatial omics, thereby enhancing our holistic understanding of lung disease biology.

Our analyses of diverse lung diseases leveraging uniLUNG have identified distinct cell subpopulations, Lym-monocytes and T-like B cells, which display distinct characteristics and preferentially emerge in specific lung pathological conditions. Previous studies have found that CD3+CD20+ T cells, this T/B dual phenotype lymphocyte, play a role in various diseases.^{98,99} Another type of CD3+CD19+ T lymphocytes, which has similar characteristics to T-like B cells, has been found in diabetes patients.¹⁰⁰ Some studies suggest that they may originate from Pro/Pre-B cells,¹⁰¹ providing support for the idea that T-like B cells may be a new B cell subset with T cell characteristics. Our analysis indicates that lymphoid monocytes increase in PF, CWP, IPAF, and sarcoidosis patients, accompanied by a decrease in CD14 monocytes and an increase in CD16 monocytes, suggesting a connection to the decline in lung function,^{102,103} exacerbated inflammation and activation of CD4 T cells.^{104,105} In LUAD and COVID-19 patients, we found that the emergence of T-like B cells is associated with T cell exhaustion, potentially serving a compensatory role. T-like B cells are also notably concentrated in CF, COPD, and ILD patients, suggesting the presence of T cell exhaustion and a similar mitochondrial function repair mechanism in these conditions. Previous studies have confirmed the presence of exhausted T cells in COPD¹⁰⁶ and ILD¹⁰⁷ patients. Importantly, we validated the unique characteristics of Lym-monocytes and T-like B cells through comparisons with several existing cell atlases (HLCA, LuCA, Pan-cancer B cell atlas and Pan-cancer T cell atlas), confirming their classification as distinct populations, especially T-like B cells, which represent a transcriptionally unique population that does not fully

align with canonical T or B cell lineages. By integrating data from various lung disease studies, our findings highlight common features across different lung diseases, aiding in establishing connections between them. Even though these findings still require further verification, they may still lay a preliminary molecular foundation for applying existing therapies targeted at specific diseases to others, or for developing new treatments applicable to multiple lung diseases.

Additionally, we identified a critical transitional malignant cell subpopulation, termed NSCLC-like SCLC cell, which is involved in the transition from NSCLC to SCLC. By integrating spatial transcriptomics, we propose that the emergence of NSCLC-like SCLC cells may rely on a specific tumour immune microenvironmental niche, co-defined by ECM remodelling, CAF activation, and spatially variable immune contexture. This subpopulation demonstrates intermediate proliferation and differentiation capabilities between mature NSCLC and SCLC cells, potentially representing a transitional malignant state during the conversion from NSCLC to SCLC, and is associated with acquired EGFR-TKI resistance.^{71–73} We detected the NSCLC-like SCLC signature in spatial transcriptomic samples from LUAD and LUSC, suggesting an increased risk of SCLC transformation in these cases. The regions enriched with NSCLC-like SCLC characteristics exhibited consistent co-localisation of CAF markers and ECM components. These findings suggest that CAFs in these regions may secrete ECM ligands that engage ITGB1 to activate EMT and promote tumour cell proliferation, migration, and stemness, ultimately creating a pro-transition microenvironment that facilitates therapy resistance.^{89,90,108,109}

We further observed subtype-specific spatial signalling associated with this population: in LUAD samples, NSCLC-like SCLC regions showed active MIF signalling from tumour to adjacent normal tissue, while in LUSC, MK signalling was more prominent. These pathways have been implicated in immune inflammation⁹¹ (MIF) and EMT activation⁹⁰ (MK), respectively, suggesting distinct signalling mechanisms that shape the tumour microenvironment to support malignant plasticity in LUAD and LUSC contexts. These pathways respectively accelerate the immune inflammatory response (MIF) and EMT (MK), as well as activate downstream pathways such as ERK/AMPK/AKT¹¹⁰ and JAK/STAT/MAPK pathways⁹² to promote the metastasis and progression of cancer cells, suggesting distinct signalling mechanisms that shape the tumour

subpopulations in LUAD on intercellular dependence was shown. **(b)** The left panels depict sender effect analysis for the tumour cell-CD4 T cell axis in LUAD samples with high NSCLC-like SCLC scores and the tumour cell-cDC1 axis in LUAD samples with low NSCLC-like SCLC scores. The y-axis represents the sender cell type, while the x-axis shows the estimated fold change in recipient cell gene expression. The right panels display sender similarity analysis, calculated based on the correlation between the coefficient vector of each sender cell type and the CD4 T cell receptor during NSCLC-like SCLC feature changes.

microenvironment to support malignant plasticity in LUAD and LUSC contexts. Furthermore, this subset is associated with increased lymphocyte infiltration in NSCLC tumours and the formation of tertiary lymphoid structures (TLS) in LUAD. While TLS generally indicate a favourable prognosis in cancers,¹¹¹ including NSCLC,⁹⁵ alterations in their composition or spatial organisation may correlate with poor outcomes,^{96,112} warranting further investigation into the specific impacts of NSCLC-like SCLC-driven TLS formation in LUAD. Cell dependency analysis suggests that NSCLC-like SCLC may cause greater structural disruption in LUAD compared to LUSC, a finding supported by sender similarity analysis. In summary, the NSCLC-like SCLC subpopulation likely represents an intermediate state during NSCLC-to-SCLC transition, broadly influencing the remodelling of the NSCLC tumour spatial microenvironment and may be related to distinct immune niche formations in LUAD and LUSC. Collectively, these observations that the characteristic regions of NSCLC-like SCLC may have a specific tumour immune microenvironment niche. This niche mainly includes the following features: the EMT process mediated by CAF-ECM-ITGB1 promotes tumour cell proliferation, metastasis and acquisition of drug resistance; spatially variable signalling pathway are involved in the microenvironment shaping process of NSCLC-to-SCLC; and local immune infiltration enhancement, especially the formation of TLS in LUAD, may produce immune exclusion or immune activation effects. Although the existing spatial analysis cannot truly provide direct evidence of the existence of the NSCLC-like SCLC cells at the single-cell resolution, further verification with higher-resolution data is needed. However, these insights may assist in developing SCLC transformation warning systems for NSCLC patients and lay the groundwork for new treatment strategies against therapy-resistant NSCLC arising from SCLC transformation.

Despite offering greater diversity and wider coverage, uniLUNG is hindered by limitations of current single-cell studies in population sampling, disease spectrum, and biological insights. For instance, its donors are mainly from European (37.5%) and Asian (45.7%) groups, lacking comprehensive population representation. About 90.3% of uniLUNG data is from healthy individuals and those with lung cancer, ILD, or COVID-19, with limited data from other diseases. A further limitation is that many pathological cell subsets remain rare and lack consensus definitions across studies, and thus have not yet been systematically incorporated into uHAF. Integrating large-scale atlases has computational challenges due to batch effects from experimental design variations. Batch correction of large matrices is costly in computing, while analysis without batch correction introduces non-biological biases. Moreover, the use of uniform QC thresholds (e.g., mitochondrial cutoffs) across diverse datasets may bias

the retention or removal of specific cell populations, and automated doublet detection and ambient RNA correction could introduce unintended side effects, potentially affecting the annotation of rare or ambiguous subsets. uniLUNG's quality relies on integration strategies, affecting its capture of cell subtype heterogeneity, seen in the unclassified cell types in uHAF. To strengthen uniLUNG's capacity to comprehensively characterise human lung cells, we will continuously supplement it with new single-cell transcriptomic and multimodal data, refine cell type annotations, and systematically incorporate pathological subsets as more high-quality disease datasets and standardised nomenclature become available.

Initially, this study established the most comprehensive unified human lung single-cell reference atlas and an interactive portal, enabling systematic exploration of lung cellular diversity and phenotypic variation across health and disease. The core atlas can be directly used as a high-quality reference or research dataset for downstream in-depth analysis, while the unverified extended atlases can be used for preliminary discovery and hypothesis generation. Leveraging this resource, we identified previously unrecognised cell subtypes across multiple lung diseases. Integrating the atlas with spatial transcriptomics data, we traced the evolution of malignant cells during lung cancer transformation and uncovered their multifaceted impact on tumour spatial microenvironment remodelling. These findings underscore the value of a unified lung cell atlas in advancing single-cell research in lung diseases and highlight the immense potential of combining spatial multi-omics data to decode lung diseases across scales.

Contributors

W.L. and X. Z. designed this study. F.W., W.C., Y.H., D.Z., and R.W. completed the data collection and curation. F.W. performed data processing and analysis. W.L., F.W., W.C., and H.T. discussed data. S.Z. and F.W. constructed the database and web platform for the atlas. Y.C., M.Y., R.Y., A.C., and J.L. assisted. H.Z. participated project management. The manuscript was written by W.L. and F.W., and W.C., H.T., H.Z., and X.Z. participated in the review and editing. All authors read and approved the final version of the manuscript. W.L., F.W., and S.Z. have accessed the verified the underlying data.

Data sharing statement

The data that support the findings of this study can be found in uniLUNG (<https://lung.unifiedcellatlas.org>) and the Zenodo repository (DOI: [10.5281/zenodo.17200374](https://doi.org/10.5281/zenodo.17200374) and [10.5281/zenodo.17230639](https://doi.org/10.5281/zenodo.17230639)), and all the originally published data are also available from the GEO repository with the following codes in [Supplementary Table S1](#) or 'Reference' of uniLUNG. Codes for this study can be obtained at <https://github.com/wufanjie97/uniLUNG>. For requests and inquiries regarding data sharing, please send an email to F.W. (wufj5@mail2.sysu.edu.cn) or W.L. (liweizhong@mail.sysu.edu.cn).

Declaration of interests

The authors declare that they have no competing interests.

Acknowledgements

This publication is part of the Human Cell Atlas—www.humancellatlas.org/publications/. The graphical abstract was created with <https://www.biorender.com/>. During the preparation of this work,

the authors used ChatGPT and DouBao in order to proofread the manuscript. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication. This work was supported by the grants of National Key R&D Program of China (2021YFF1200900 and 2021YFF1200903), National Natural Science Foundation of China (92474107), Guangdong Basic and Applied Basic Research Foundation of China (2022B1515120077), Major Project of Guangzhou National Laboratory of China (GZNL2024A01003), and Support Scheme of Guangzhou for Leading Talents in Innovation and Entrepreneurship of China (2020007).

Appendix A. Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.jbiom.2025.106025>.

References

- Ardini-Poleske ME, Clark RF, Ansong C, et al. LungMAP: the molecular atlas of lung development program. *Am J Physiol Lung Cell Mol Physiol*. 2017;313(5):L733–L740.
- Sauler M, McDonough JE, Adams TS, et al. Characterization of the COPD alveolar niche using single-cell RNA sequencing. *Nat Commun*. 2022;13(1):494.
- Sun X, Perl AK, Li R, et al. A census of the lung: CellCards from LungMAP. *Dev Cell*. 2022;57(1):112–145e2.
- Travaglini KJ, Nabhan AN, Penland L, et al. A molecular cell atlas of the human lung from single-cell RNA sequencing. *Nature*. 2020;587(7835):619–625.
- Vieira Braga FA, Kar G, Berg M, et al. A cellular census of human lungs identifies novel cell states in health and in asthma. *Nat Med*. 2019;25(7):1153–1163.
- Enfield KSS, Colliver E, Lee C, et al. Spatial architecture of myeloid and T cells orchestrates immune evasion and clinical outcome in lung cancer. *Cancer Discov*. 2024;14(6):1018–1047.
- Vannan A, Lyu R, Williams AL, et al. Spatial transcriptomics identifies molecular niche dysregulation associated with distal lung remodeling in pulmonary fibrosis. *Nat Genet*. 2025;57(3):647–658.
- Wang Y, Liu B, Min Q, et al. Spatial transcriptomics delineates molecular features and cellular plasticity in lung adenocarcinoma progression. *Cell Discov*. 2023;9(1):96.
- Gulati GS, D'Silva JP, Liu Y, Wang L, Newman AM. Profiling cell identity and tissue architecture with single-cell and spatial transcriptomics. *Nat Rev Mol Cell Biol*. 2025;26(1):11–31.
- Mitamura Y, Reiger M, Kim J, et al. Spatial transcriptomics combined with single-cell RNA-sequencing unravels the complex inflammatory cell network in atopic dermatitis. *Allergy*. 2023;78(8):2215–2231.
- Zhang L, Chen D, Song D, et al. Clinical and translational values of spatial transcriptomics. *Signal Transduct Target Ther*. 2022;7(1):111.
- Deprez M, Zaragosi LE, Truchi M, et al. A single-cell atlas of the human healthy airways. *Am J Respir Crit Care Med*. 2020;202(12):1636–1645.
- Madisson E, Oliver AJ, Kleshchevnikov V, et al. A spatially resolved atlas of the human lung characterizes a gland-associated immune niche. *Nat Genet*. 2023;55(1):66–77.
- Wang Z, Li Z, Zhou K, et al. Deciphering cell lineage specification of human lung adenocarcinoma with single-cell RNA sequencing. *Nat Commun*. 2021;12(1):6500.
- Han XP, Zhou ZM, Fei LJ, et al. Construction of a human cell landscape at single-cell level. *Nature*. 2020;581(7808):303–309.
- Schupp JC, Adams TS, Cosme C Jr, et al. Integrated single-cell atlas of endothelial cells of the human lung. *Circulation*. 2021;144(4):286–302.
- Sikkema L, Ramirez-Suastegui C, Strobl DC, et al. An integrated cell atlas of the lung in health and disease. *Nat Med*. 2023;29(6):1563–1577.
- Barrett T, Wilhite SE, Ledoux P, et al. NCBI GEO: archive for functional genomics data sets—update. *Nucleic Acids Res*. 2013;41(Database issue):D991–D995.
- Chan JM, Quintana-Villalonga A, Gao VR, et al. Signatures of plasticity, metastasis, and immunosuppression in an atlas of human small cell lung cancer. *Cancer Cell*. 2021;39(11):1479–1496e18.
- Li M, Zhang X, Ang KS, et al. DISCO: a database of Deeply Integrated human Single-Cell Omics data. *Nucleic Acids Res*. 2022;50(D1):D596–D602.
- Moreno P, Huang N, Manning JR, et al. User-friendly, scalable tools and workflows for single-cell RNA-seq analysis. *Nat Methods*. 2021;18(4):327–328.
- Regev A, Teichmann SA, Lander ES, et al. The human cell atlas. *Elife*. 2017;6:e27041.
- Salcher S, Sturm G, Horvath L, et al. High-resolution single-cell atlas reveals diversity and plasticity of tissue-resident neutrophils in non-small cell lung cancer. *Cancer Cell*. 2022;40(12):1503–1520e8.
- Snyder MP, Lin S, Posgai A, et al. The human body at cellular resolution: the NIH Human Biomolecular Atlas Program. *Nature*. 2019;574(7777):187–192.
- Speir ML, Bhaduri A, Markov NS, et al. UCSC Cell Browser: visualize your single-cell data. *Bioinformatics*. 2021;37(23):4578–4580.
- Sun D, Wang J, Han Y, et al. TISCH: a comprehensive web resource enabling interactive single-cell transcriptome visualization of tumor microenvironment. *Nucleic Acids Res*. 2021;49(D1):D1420–D1430.
- Yuan H, Yan M, Zhang G, et al. CancerSEA: a cancer single-cell state atlas. *Nucleic Acids Res*. 2019;47(D1):D900–D908.
- Zeng JY, Zhang YD, Shang YF, et al. CancerSCEM: a database of single-cell expression map across various human cancers. *Nucleic Acids Res*. 2022;50(D1):D1147–D1155.
- Hao Y, Hao S, Andersen-Nissen E, et al. Integrated analysis of multimodal single-cell data. *Cell*. 2021;184(13):3573–3587e29.
- Zilionis R, Engblom C, Pfirschke C, et al. Single-cell transcriptomics of human and mouse lung cancers reveals conserved myeloid populations across individuals and species. *Immunity*. 2019;50(5):1317–1334e10.
- Yang S, Corbett SE, Koga Y, et al. Decontamination of ambient RNA in single-cell RNA-seq with DecontX. *Genome Biol*. 2020;21(1):57.
- McGinnis CS, Murrow LM, Gartner ZJ. DoubletFinder: doublet detection in single-cell RNA sequencing data using artificial nearest neighbors. *Cell Syst*. 2019;8(4):329–337e4.
- Xu C, Prete M, Webb S, et al. Automatic cell-type harmonization and integration across Human Cell Atlas datasets. *Cell*. 2023;186(26):5876–5891e20.
- Johnson WE, Li C, Rabinovic A. Adjusting batch effects in microarray expression data using empirical Bayes methods. *Biostatistics*. 2007;8(1):118–127.
- Korsunsky I, Millard N, Fan J, et al. Fast, sensitive and accurate integration of single-cell data with Harmony. *Nat Methods*. 2019;16(12):1289–1296.
- Hie B, Bryson B, Berger B. Efficient integration of heterogeneous single-cell transcriptomes using Scanorama. *Nat Biotechnol*. 2019;37(6):685–691.
- Lopez R, Regier J, Cole MB, Jordan MI, Yosef N. Deep generative modeling for single-cell transcriptomics. *Nat Methods*. 2018;15(12):1053–1058.
- Xu C, Lopez R, Mehlman E, Regier J, Jordan MI, Yosef N. Probabilistic harmonization and annotation of single-cell transcriptomics data with deep generative models. *Mol Syst Biol*. 2021;17(1):e9620.
- Polanski K, Young MD, Miao Z, Meyer KB, Teichmann SA, Park JE. BBKNN: fast batch alignment of single cell transcriptomes. *Bioinformatics*. 2020;36(3):964–965.
- Welch JD, Kozareva V, Ferreira A, Vanderburg C, Martin C, Macosko EZ. Single-cell multi-omic integration compares and contrasts features of brain cell identity. *Cell*. 2019;177(7):1873–1887e17.
- Hao Y, Stuart T, Kowalski MH, et al. Dictionary learning for integrative, multimodal and scalable single-cell analysis. *Nat Biotechnol*. 2024;42(2):293–304.
- Haghverdi L, Lun ATL, Morgan MD, Marioni JC. Batch effects in single-cell RNA-sequencing data are corrected by matching mutual nearest neighbors. *Nat Biotechnol*. 2018;36(5):421–427.
- Lueken MD, Buttner M, Chaichoompu K, et al. Benchmarking atlas-level data integration in single-cell genomics. *Nat Methods*. 2022;19(1):41–50.
- Dominguez Conde C, Xu C, Jarvis LB, et al. Cross-tissue immune cell analysis reveals tissue-specific features in humans. *Science*. 2022;376(6594):eab15197.

- 45 Buttner M, Ostner J, Muller CL, Theis FJ, Schubert B. scCODA is a Bayesian model for compositional single-cell data analysis. *Nat Commun*. 2021;12(1):6876.
- 46 Gaujoux R, Seoighe C. A flexible R package for nonnegative matrix factorization. *BMC Bioinformatics*. 2010;11:367.
- 47 Zheng L, Qin S, Si W, et al. Pan-cancer single-cell landscape of tumor-infiltrating T cells. *Science*. 2021;374(6574):abe6474.
- 48 Squair JW, Gautier M, Kathe C, et al. Confronting false discoveries in single-cell differential expression. *Nat Commun*. 2021;12(1):5692.
- 49 Robinson MD, McCarthy DJ, Smyth GK. edgeR: a Bioconductor package for differential expression analysis of digital gene expression data. *Bioinformatics*. 2010;26(1):139–140.
- 50 Wu T, Hu E, Xu S, et al. clusterProfiler 4.0: a universal enrichment tool for interpreting omics data. *Innovation (Camb)*. 2021;2(3):100141.
- 51 Wu H, Zhao X, Hochrein SM, et al. Mitochondrial dysfunction promotes the transition of precursor to terminally exhausted T cells through HIF-1 α -mediated glycolytic reprogramming. *Nat Commun*. 2023;14(1):6858.
- 52 Tang Z, Kang B, Li C, Chen T, Zhang Z. GEPIA2: an enhanced web server for large-scale expression profiling and interactive analysis. *Nucleic Acids Res*. 2019;47(W1):W556–W560.
- 53 Aibar S, Gonzalez-Blas CB, Moerman T, et al. SCENIC: single-cell regulatory network inference and clustering. *Nat Methods*. 2017;14(11):1083–1086.
- 54 Lotfollahi M, Naghipourfar M, Luecken MD, et al. Mapping single-cell data to reference atlases by transfer learning. *Nat Biotechnol*. 2022;40(1):121–130.
- 55 Ma J, Wu Y, Ma L, et al. A blueprint for tumor-infiltrating B cells across human cancers. *Science*. 2024;384(6695):eadj4857.
- 56 Patel AP, Tirosh I, Trombetta JJ, et al. Single-cell RNA-seq highlights intratumoral heterogeneity in primary glioblastoma. *Science*. 2014;344(6190):1396–1401.
- 57 Peng J, Sun BF, Chen CY, et al. Single-cell RNA-seq highlights intra-tumoral heterogeneity and malignant progression in pancreatic ductal adenocarcinoma. *Cell Res*. 2019;29(9):725–738.
- 58 Gulati GS, Sikandar SS, Wesche DJ, et al. Single-cell transcriptional diversity is a hallmark of developmental potential. *Science*. 2020;367(6476):405–411.
- 59 Cao J, Spielmann M, Qiu X, et al. The single-cell transcriptional landscape of mammalian organogenesis. *Nature*. 2019;566(7745):496–502.
- 60 Wolf FA, Angerer P, Theis FJ. SCANPY: large-scale single-cell gene expression data analysis. *Genome Biol*. 2018;19(1):15.
- 61 Xu H, Fu H, Long Y, et al. Unsupervised spatially embedded deep representation of spatial transcriptomics. *Genome Med*. 2024;16(1):12.
- 62 Zhong Z, Hou J, Yao Z, et al. Domain generalization enables general cancer cell annotation in single-cell and spatial transcriptomics. *Nat Commun*. 2024;15(1):1929.
- 63 Liu Q, Hsu CY, Shyr Y. Scalable and model-free detection of spatial patterns and colocalization. *Genome Res*. 2022;32(9):1736–1745.
- 64 Cang Z, Zhao Y, Almet AA, et al. Screening cell-cell communication in spatial transcriptomics via collective optimal transport. *Nat Methods*. 2023;20(2):218–228.
- 65 Fischer DS, Schaar AC, Theis FJ. Modeling intercellular communication in tissues using spatial graphs of cells. *Nat Biotechnol*. 2023;41(3):332–336.
- 66 Jiang S, Qian Q, Zhu T, et al. Cell Taxonomy: a curated repository of cell types with multifaceted characterization. *Nucleic Acids Res*. 2023;51(D1):D853–D860.
- 67 Hu C, Li T, Xu Y, et al. CellMarker 2.0: an updated database of manually curated cell markers in human/mouse and web tools based on scRNA-seq data. *Nucleic Acids Res*. 2023;51(D1):D870–D876.
- 68 Chen S, Luo Y, Gao H, et al. hECA: the cell-centric assembly of a cell atlas. *iScience*. 2022;25(5):104318.
- 69 Wilk AJ, Rustagi A, Zhao NQ, et al. A single-cell atlas of the peripheral immune response in patients with severe COVID-19. *Nat Med*. 2020;26(7):1070–1076.
- 70 Guo X, Zhang Y, Zheng L, et al. Global characterization of T cells in non-small-cell lung cancer by single-cell sequencing. *Nat Med*. 2018;24(7):978–985.
- 71 Lee JK, Lee J, Kim S, et al. Clonal history and genetic predictors of transformation into small-cell carcinomas from lung adenocarcinomas. *J Clin Oncol*. 2017;35(26):3065–3074.
- 72 Li Y, Xie T, Wang S, et al. Mechanism exploration and model construction for small cell transformation in EGFR-mutant lung adenocarcinomas. *Signal Transduct Target Ther*. 2024;9(1):261.
- 73 Oser MG, Niederst MJ, Sequist LV, Engelman JA. Transformation from non-small-cell lung cancer to small-cell lung cancer: molecular drivers and cells of origin. *Lancet Oncol*. 2015;16(4):e165–e172.
- 74 Tong X, Patel AS, Kim E, et al. Adeno-to-squamous transition drives resistance to KRAS inhibition in LKB1 mutant lung cancer. *Cancer Cell*. 2024;42(3):413–428e7.
- 75 Awad MM, Liu S, Rybkin II, et al. Acquired resistance to KRAS (G12C) inhibition in cancer. *N Engl J Med*. 2021;384(25):2382–2393.
- 76 Offin M, Chan JM, Tenet M, et al. Concurrent RB1 and TP53 alterations define a subset of EGFR-mutant lung cancers at risk for histologic transformation and inferior clinical outcomes. *J Thorac Oncol*. 2019;14(10):1784–1793.
- 77 Chen J, Larsson L, Swarbrick A, Lundeborg J. Spatial landscapes of cancers: insights and opportunities. *Nat Rev Clin Oncol*. 2024;21(9):660–674.
- 78 Gong D, Arbesfeld-Qiu JM, Perrault E, Bae JW, Hwang WL. Spatial oncology: translating contextual biology to the clinic. *Cancer Cell*. 2024;42(10):1653–1675.
- 79 Nath S, Mukherjee P. MUC1: a multifaceted oncoprotein with a key role in cancer progression. *Trends Mol Med*. 2014;20(6):332–342.
- 80 Li H, Yang T, Chen Y, Xie Z. Single cell RNA-seq and bulk RNA-seq analysis identifies MUC1 as a key gene for lung adenocarcinoma to neuroendocrine transformation. *Transl Lung Cancer Res*. 2025;14(3):824–841.
- 81 Zhen S, Jia Y, Zhao Y, et al. NEAT1_1 confers gefitinib resistance in lung adenocarcinoma through promoting AKR1C1-mediated ferroptosis defence. *Cell Death Discov*. 2024;10(1):131.
- 82 Uuskula-Reimand L, Wilson MD. Untangling the roles of TOP2A and TOP2B in transcription and cancer. *Sci Adv*. 2022;8(44):eadd4920.
- 83 Loibl S, Poortmans P, Morrow M, Denkert C, Curigliano G. Breast cancer. *Lancet*. 2021;397(10286):1750–1769.
- 84 Aguirre-Portoles C, Bird AW, Hyman A, Canamero M, Perez de Castro I, Malumbres M. Tpx2 controls spindle integrity, genome stability, and tumor development. *Cancer Res*. 2012;72(6):1518–1528.
- 85 De Zuani M, Xue H, Park JS, et al. Single-cell and spatial transcriptomics analysis of non-small cell lung cancer. *Nat Commun*. 2024;15(1):4388.
- 86 Sarode P, Mansouri S, Karger A, et al. Epithelial cell plasticity defines heterogeneity in lung cancer. *Cell Signal*. 2020;65:109463.
- 87 Chen Y, Toth R, Chocarro S, et al. Club cells employ regeneration mechanisms during lung tumorigenesis. *Nat Commun*. 2022;13(1):4557.
- 88 Zhang D, Wang W, Zhou H, et al. ANXA1: an important independent prognostic factor and molecular target in glioma. *Front Genet*. 2022;13:851505.
- 89 Meder L, Orschel CI, Otto CJ, et al. Blocking the angiopoietin-2-dependent integrin beta-1 signaling axis abrogates small cell lung cancer invasion and metastasis. *JCI Insight*. 2024;9(10):e166402.
- 90 Li Y, Sun C, Tan Y, Zhang H, Li Y, Zou H. ITGB1 enhances the radioresistance of human non-small cell lung cancer cells by modulating the DNA damage response and YAP1-induced epithelial-mesenchymal transition. *Int J Biol Sci*. 2021;17(2):635–650.
- 91 Sumaiya K, Langford D, Natarajaseenivasan K, Shanmughapriya S. Macrophage migration inhibitory factor (MIF): a multifaceted cytokine regulated by genetic and physiological strategies. *Pharmacol Ther*. 2022;233:108024.
- 92 Xu C, Zhu S, Wu M, Han W, Yu Y. Functional receptors and intracellular signal pathways of midkine (MK) and pleiotrophin (PTN). *Biol Pharm Bull*. 2014;37(4):511–520.
- 93 Kleshchevnikov V, Shmatko A, Dann E, et al. Cell2location maps fine-grained cell types in spatial transcriptomics. *Nat Biotechnol*. 2022;40(5):661–671.
- 94 Lauss M, Donia M, Svane IM, Jonsson G. B cells and tertiary lymphoid structures: friends or foes in cancer immunotherapy? *Clin Cancer Res*. 2022;28(9):1751–1758.
- 95 Dieu-Nosjean MC, Goc J, Giraldo NA, Sautes-Fridman C, Fridman WH. Tertiary lymphoid structures in cancer and beyond. *Trends Immunol*. 2014;35(11):571–580.

- 96 Munoz-Erazo L, Rhodes JL, Marion VC, Kemp RA. Tertiary lymphoid structures in cancer - considerations for patient prognosis. *Cell Mol Immunol.* 2020;17(6):570–575.
- 97 Germain C, Gnjjatic S, Tamzalit F, et al. Presence of B cells in tertiary lymphoid structures is associated with a protective immunity in patients with lung cancer. *Am J Respir Crit Care Med.* 2014;189(7):832–844.
- 98 Schuh E, Berer K, Mulazzani M, et al. Features of human CD3+CD20+ T cells. *J Immunol.* 2016;197(4):1111–1117.
- 99 von Essen MR, Ammitzboll C, Hansen RH, et al. Proinflammatory CD20+ T cells in the pathogenesis of multiple sclerosis. *Brain.* 2019;142(1):120–132.
- 100 Ahmed R, Omidian Z, Giwa A, et al. A public BCR present in a unique dual-receptor-expressing lymphocyte from type 1 diabetes patients encodes a potent T cell autoantigen. *Cell.* 2019;177(6):1583–1599e16.
- 101 Zhang Y, Guo C, Zhou Y, et al. A biphenotypic lymphocyte subset displays both T- and B-cell functionalities. *Commun Biol.* 2024;7(1):28.
- 102 Fraser SD, Crooks MG, Kaye PM, Hart SP. Distinct immune regulatory receptor profiles linked to altered monocyte subsets in sarcoidosis. *ERJ Open Res.* 2021;7(1).
- 103 Bingham GC, Muehling LM, Li C, et al. High-dimensional comparison of monocytes and T cells in post-COVID and idiopathic pulmonary fibrosis. *Front Immunol.* 2023;14:1308594.
- 104 Wong KL, Yeap WH, Tai JJ, Ong SM, Dang TM, Wong SC. The three human monocyte subsets: implications for health and disease. *Immunol Res.* 2012;53(1–3):41–57.
- 105 Ozanska A, Szymczak D, Rybka J. Pattern of human monocyte subpopulations in health and disease. *Scand J Immunol.* 2020;92(1):e12883.
- 106 Kalathil SG, Lugade AA, Pradhan V, et al. T-regulatory cells and programmed death 1+ T cells contribute to effector T-cell dysfunction in patients with chronic obstructive pulmonary disease. *Am J Respir Crit Care Med.* 2014;190(1):40–50.
- 107 Fraz MSA, Michelsen AE, Moe N, et al. Raised serum markers of T cell activation and exhaustion in granulomatous-lymphocytic interstitial lung disease in common variable immunodeficiency. *J Clin Immunol.* 2022;42(7):1553–1563.
- 108 Arpinati L, Carradori G, Scherz-Shouval R. CAF-induced physical constraints controlling T cell state and localization in solid tumours. *Nat Rev Cancer.* 2024;24(10):676–693.
- 109 Mao X, Xu J, Wang W, et al. Crosstalk between cancer-associated fibroblasts and immune cells in the tumor microenvironment: new findings and future perspectives. *Mol Cancer.* 2021;20(1):131.
- 110 Jankauskas SS, Wong DWL, Bucala R, Djudjaj S, Boor P. Evolving complexity of MIF signaling. *Cell Signal.* 2019;57:76–88.
- 111 Sautes-Fridman C, Petitprez F, Calderaro J, Fridman WH. Tertiary lymphoid structures in the era of cancer immunotherapy. *Nat Rev Cancer.* 2019;19(6):307–325.
- 112 Ding GY, Ma JQ, Yun JP, et al. Distribution and density of tertiary lymphoid structures predict clinical outcome in intrahepatic cholangiocarcinoma. *J Hepatol.* 2022;76(3):608–618.